

ML Hybrid Models Comparison Report

Generated: 2025-12-05T14:56:56.618046

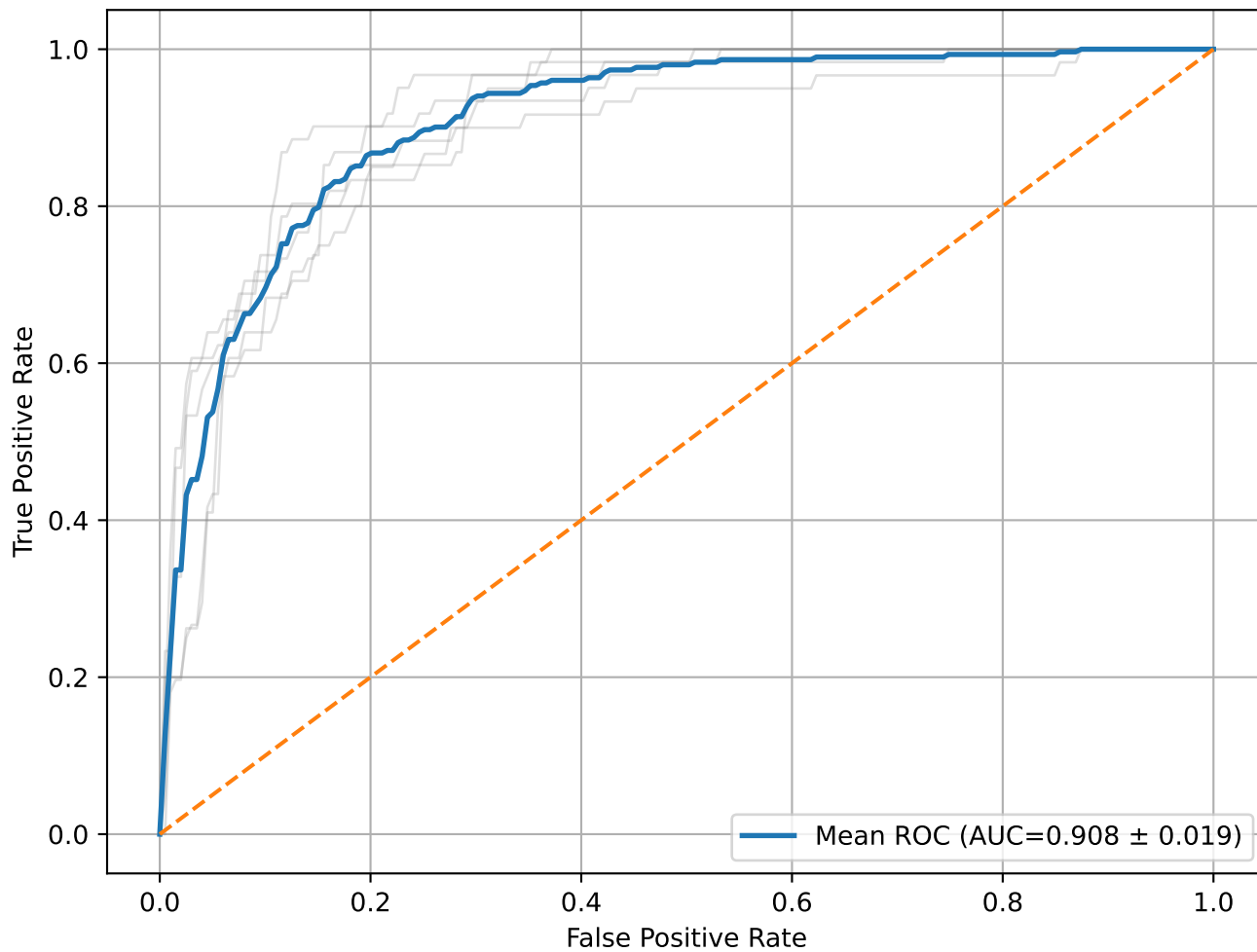
Dataset samples: 1000

Models: logistic, rf, svm, xgb, lgb, dnn, stacking, voting, bagging, pca_svm, ga_xgb, pso_svm, bayes_opt_svms

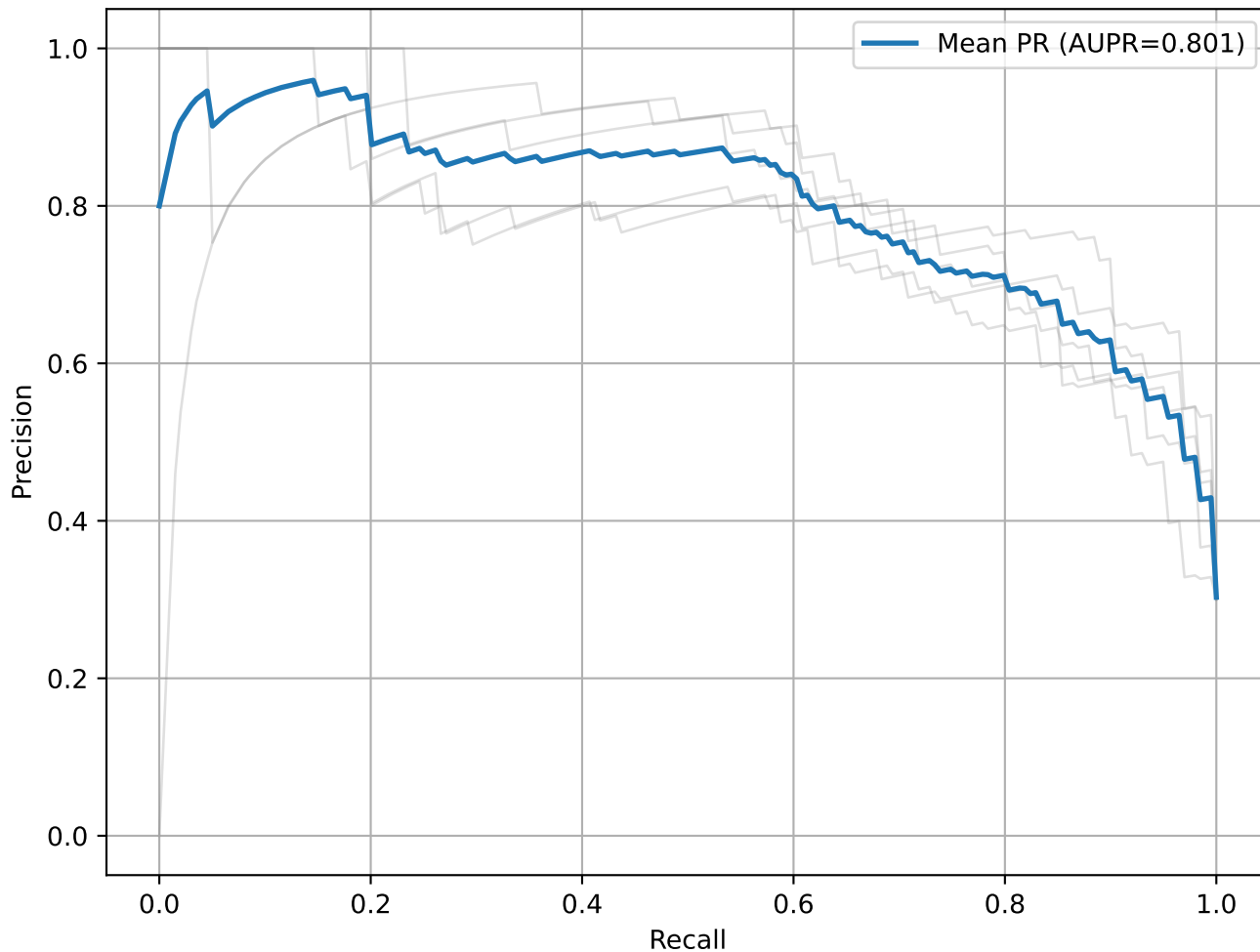
StratifiedKFold: 5 splits

Random seed: 42

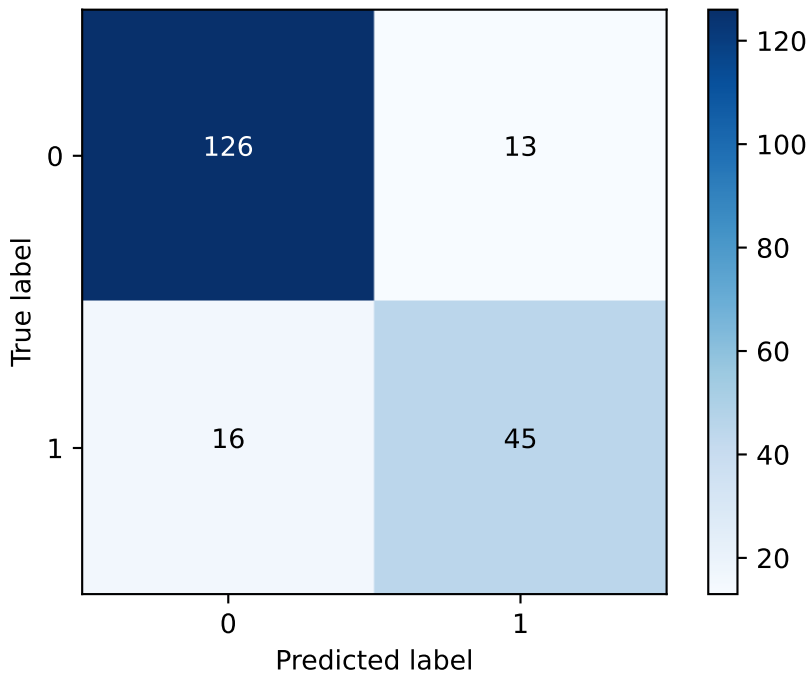
ROC - LOGISTIC



Precision-Recall - LOGISTIC



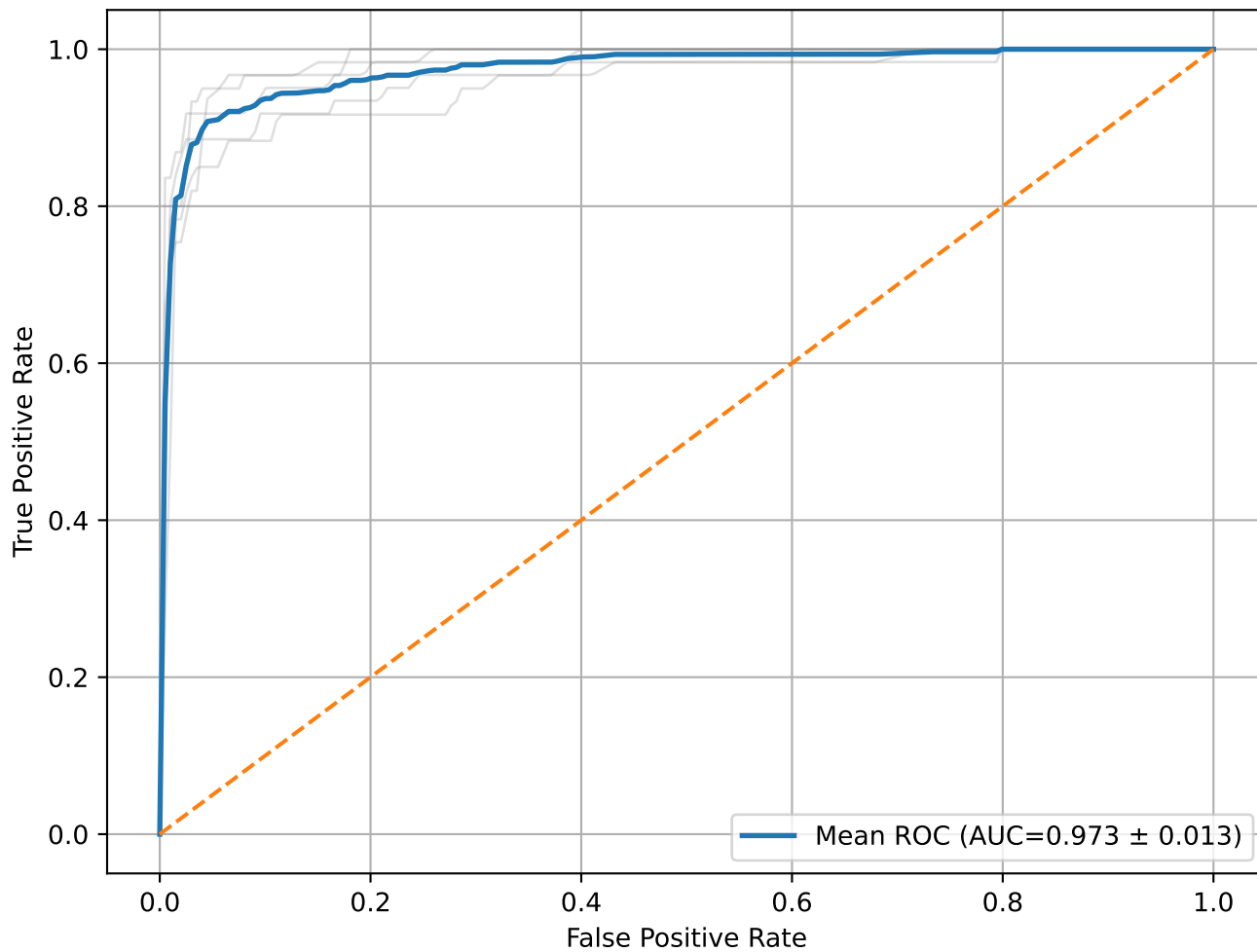
Confusion Matrix - LOGISTIC (last fold)



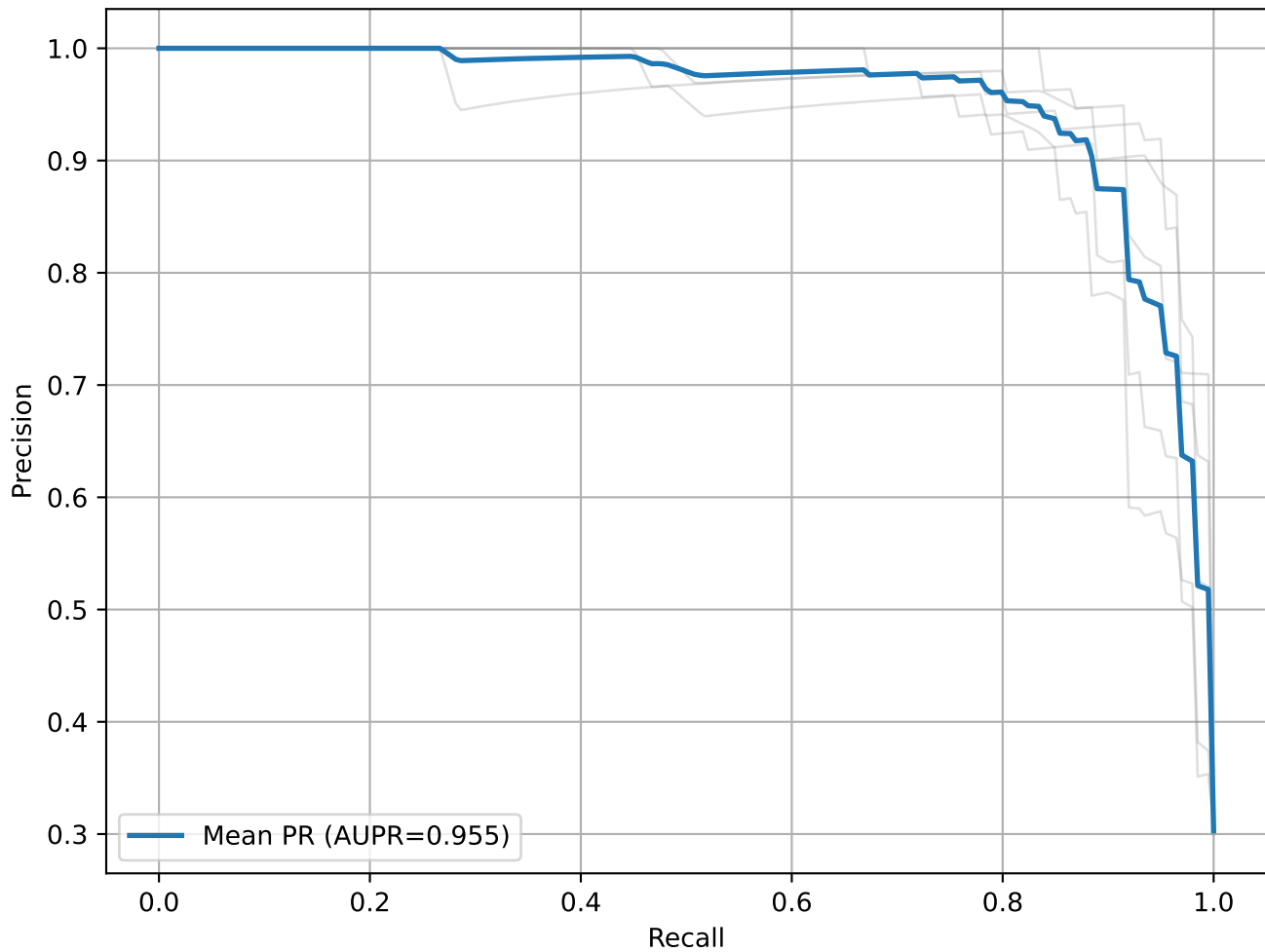
	precision	recall	f1-score	support
0	0.89	0.91	0.90	139
1	0.78	0.74	0.76	61
accuracy			0.85	200
macro avg	0.83	0.82	0.83	200
weighted avg	0.85	0.85	0.85	200

Classification Report - LOGISTIC (last fold)

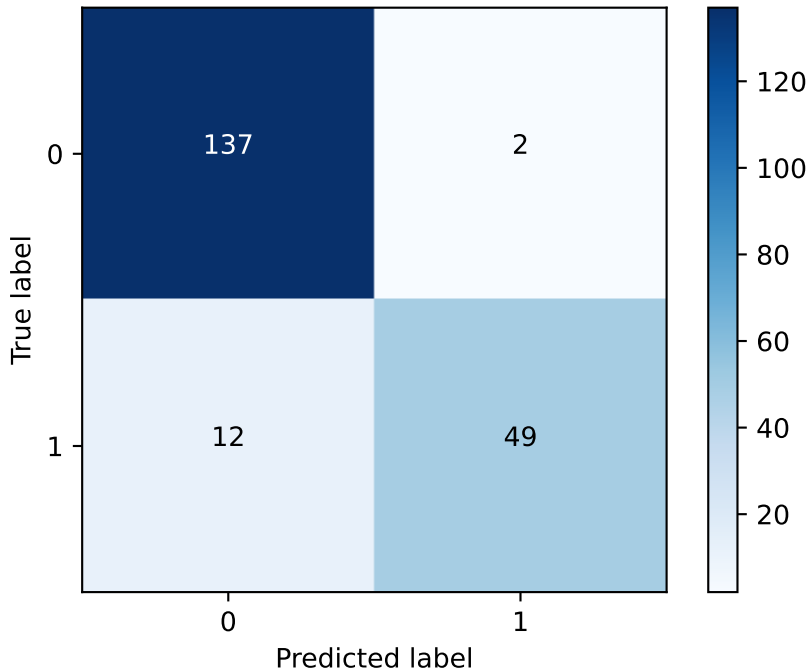
ROC - RF



Precision-Recall - RF



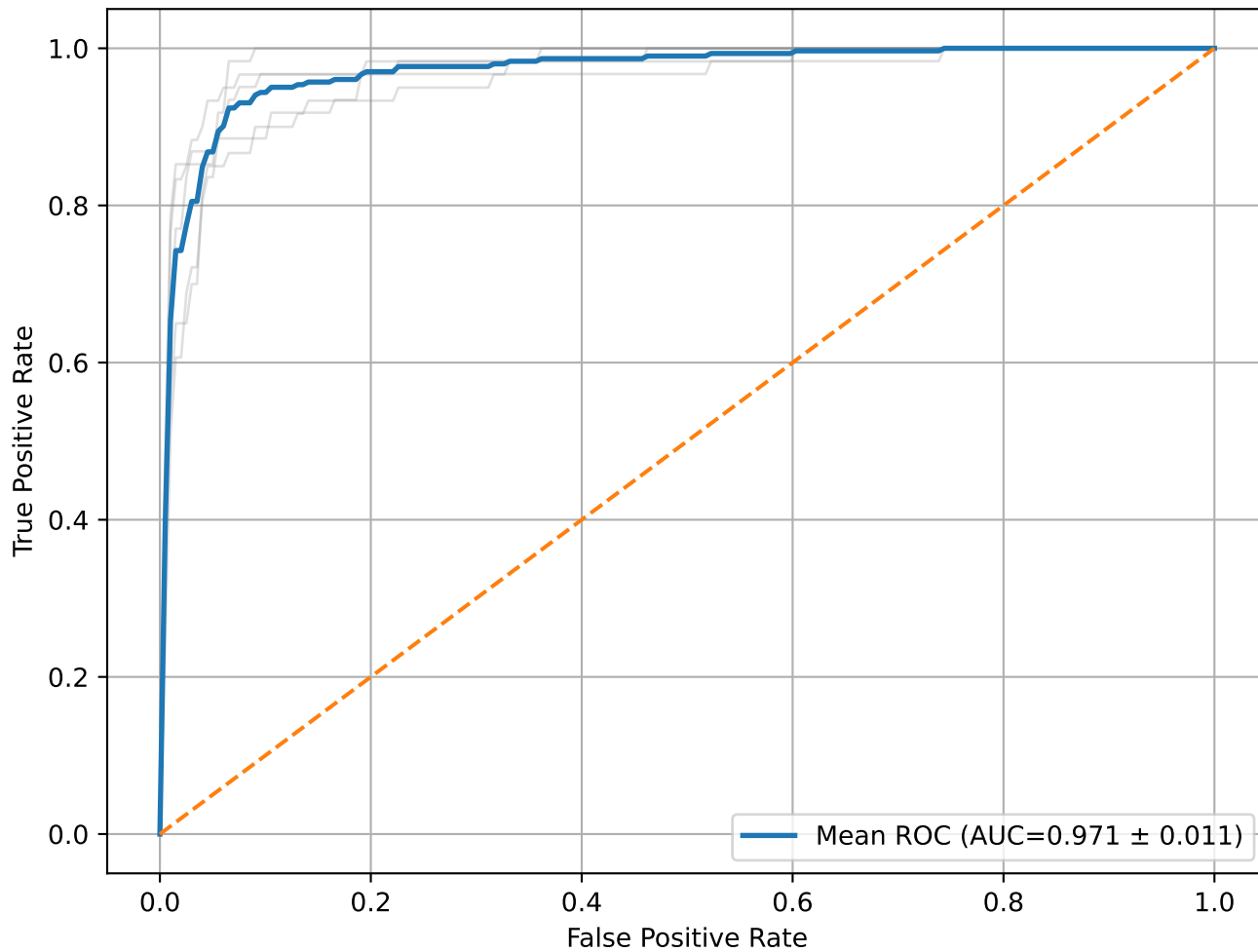
Confusion Matrix - RF (last fold)



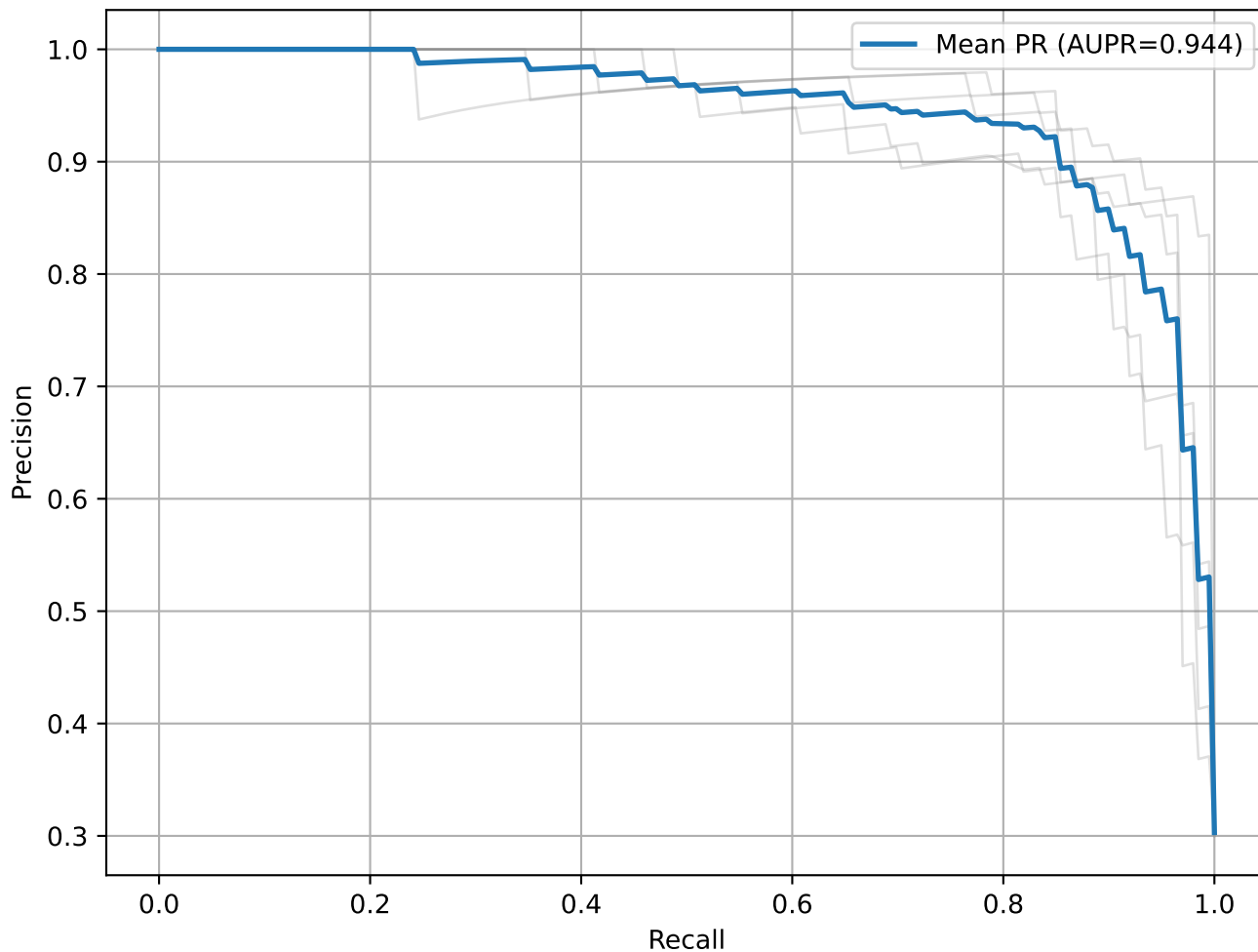
	precision	recall	f1-score	support
0	0.92	0.99	0.95	139
1	0.96	0.80	0.88	61
accuracy			0.93	200
macro avg	0.94	0.89	0.91	200
weighted avg	0.93	0.93	0.93	200

Classification Report - RF (last fold)

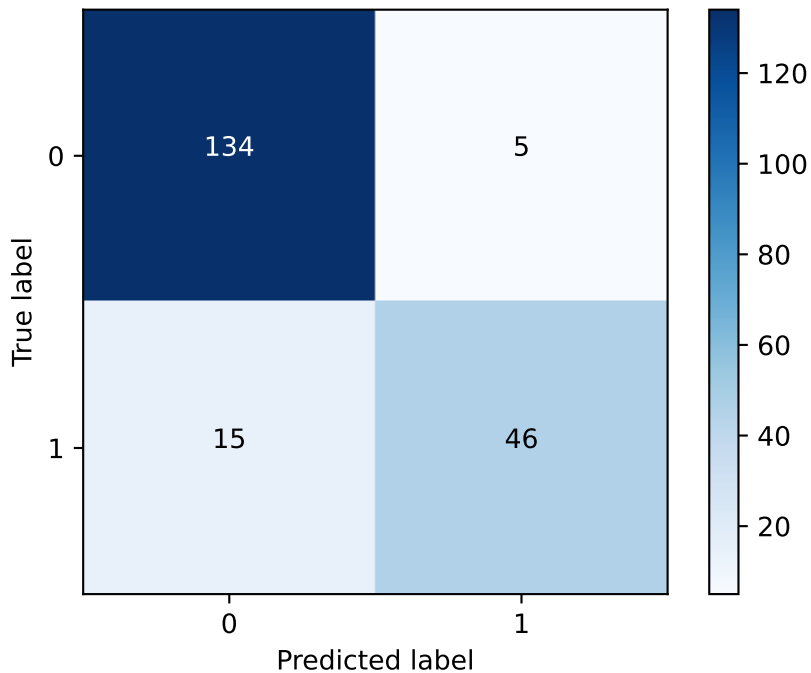
ROC - SVM



Precision-Recall - SVM



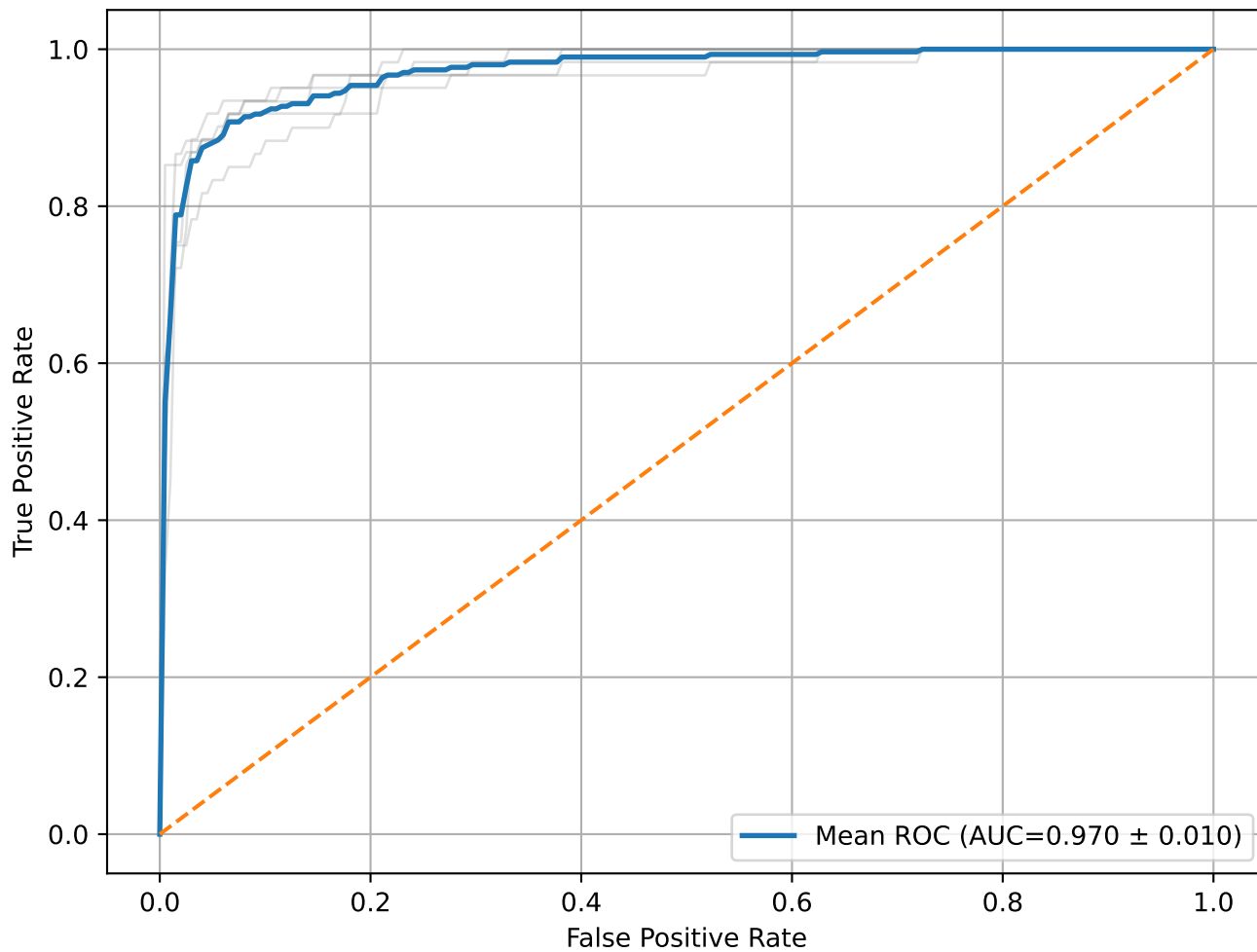
Confusion Matrix - SVM (last fold)



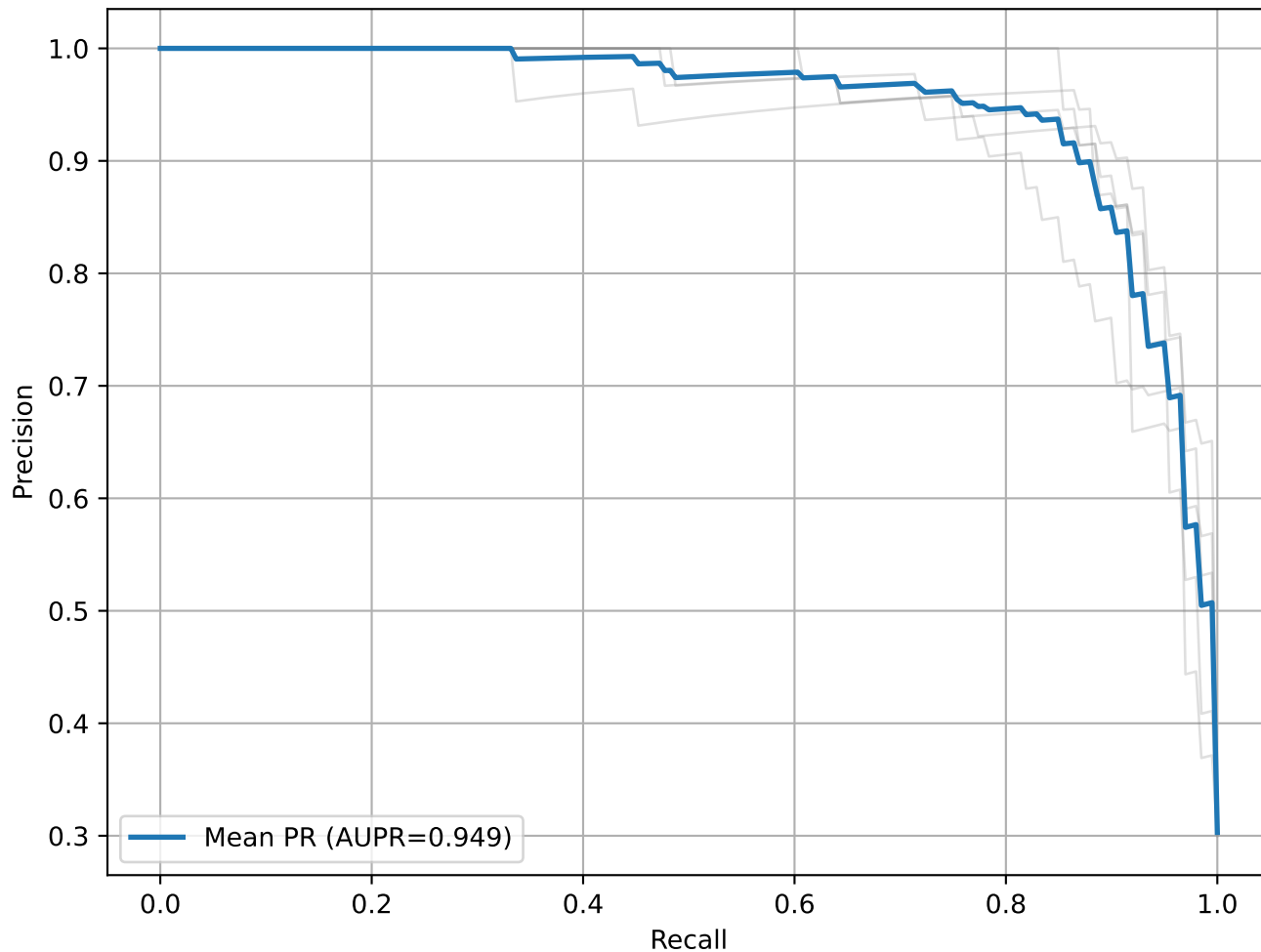
	precision	recall	f1-score	support
0	0.90	0.96	0.93	139
1	0.90	0.75	0.82	61
accuracy			0.90	200
macro avg	0.90	0.86	0.88	200
weighted avg	0.90	0.90	0.90	200

Classification Report - SVM (last fold)

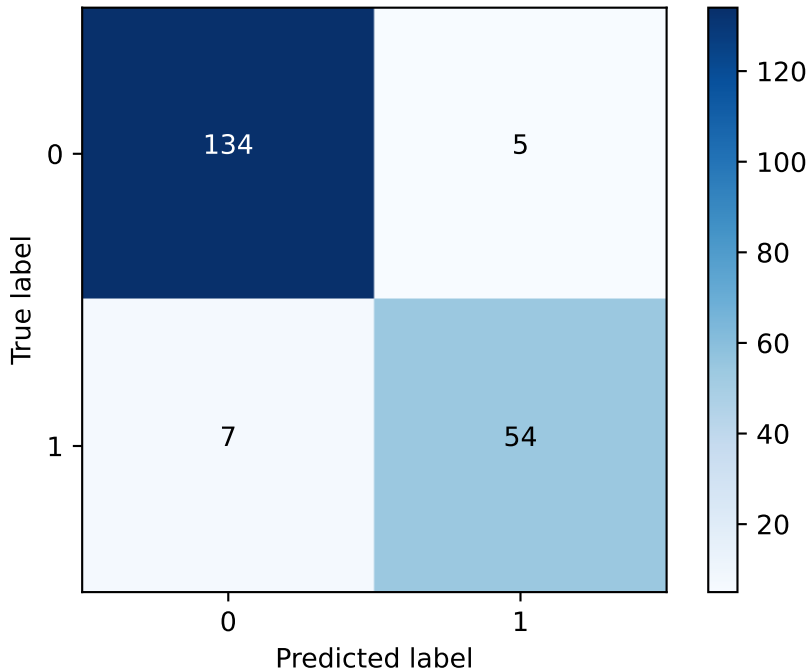
ROC - XGB



Precision-Recall - XGB



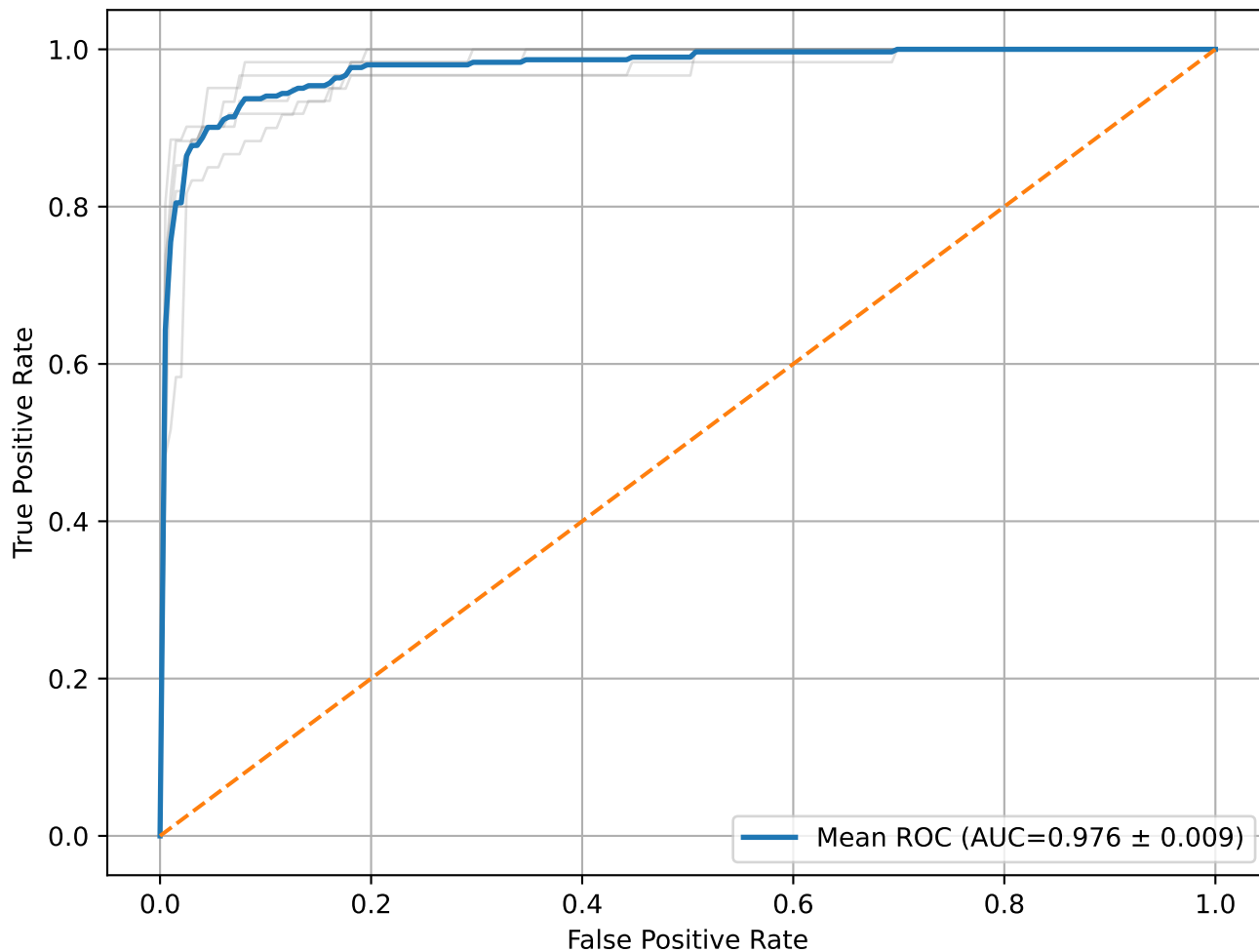
Confusion Matrix - XGB (last fold)



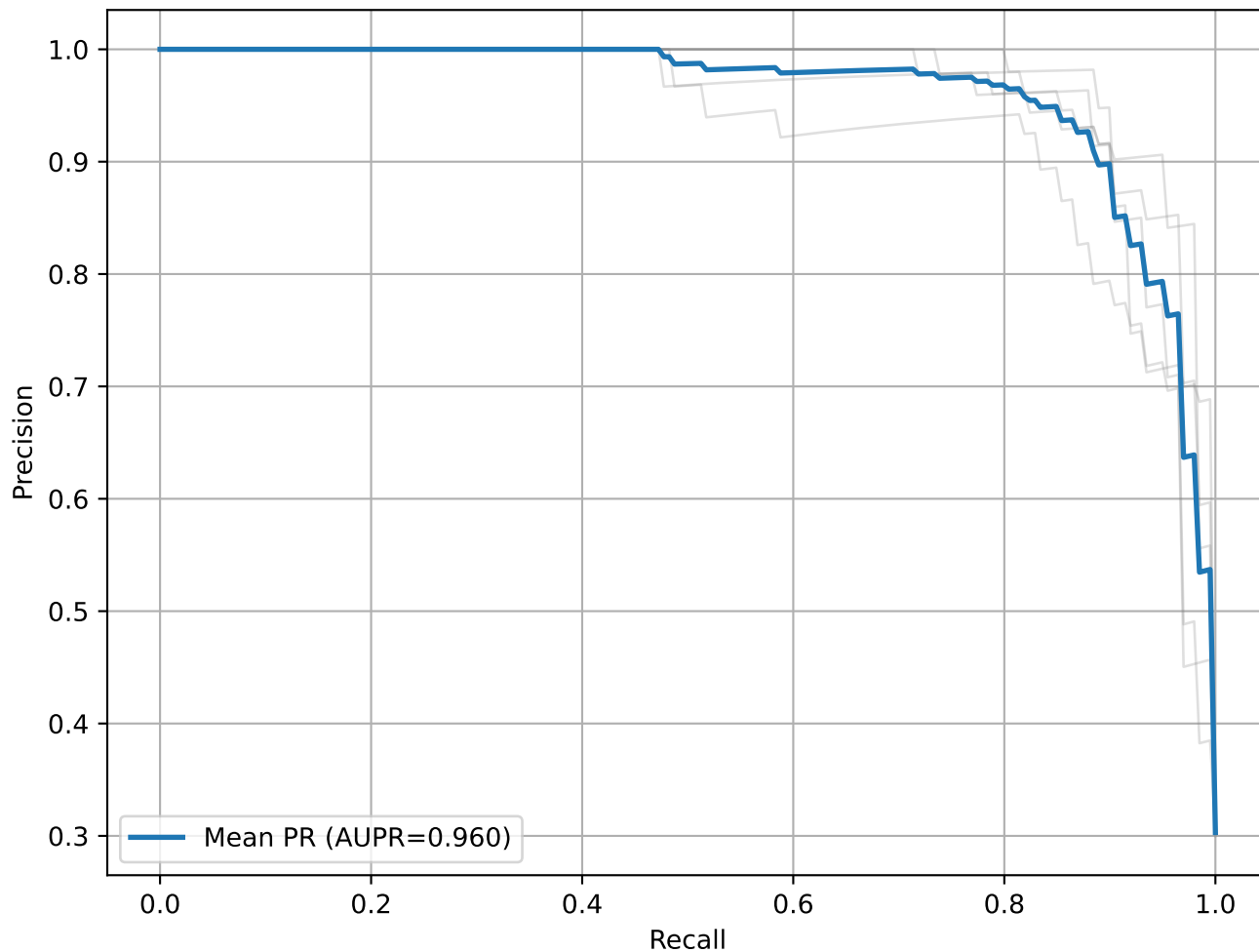
	precision	recall	f1-score	support
0	0.95	0.96	0.96	139
1	0.92	0.89	0.90	61
accuracy			0.94	200
macro avg	0.93	0.92	0.93	200
weighted avg	0.94	0.94	0.94	200

Classification Report - XGB (last fold)

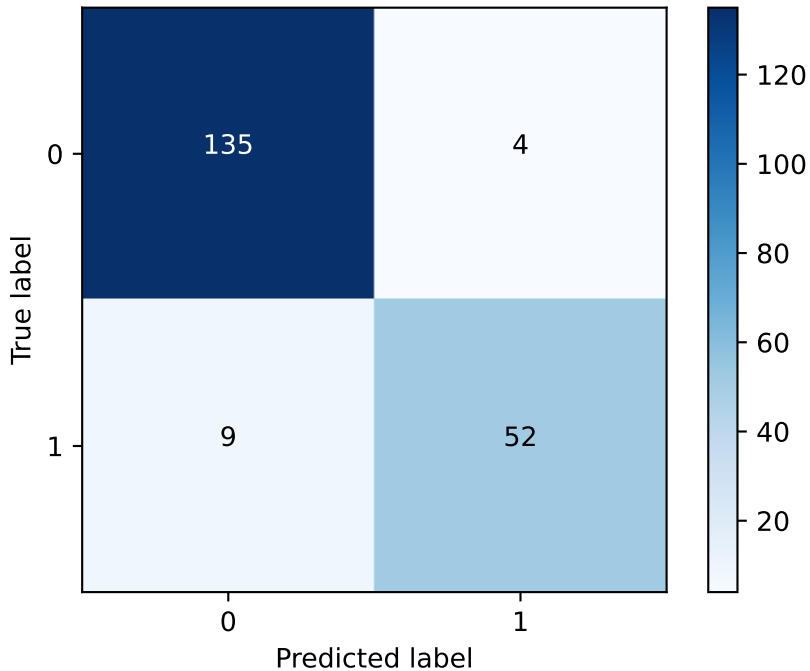
ROC - LGB



Precision-Recall - LGB



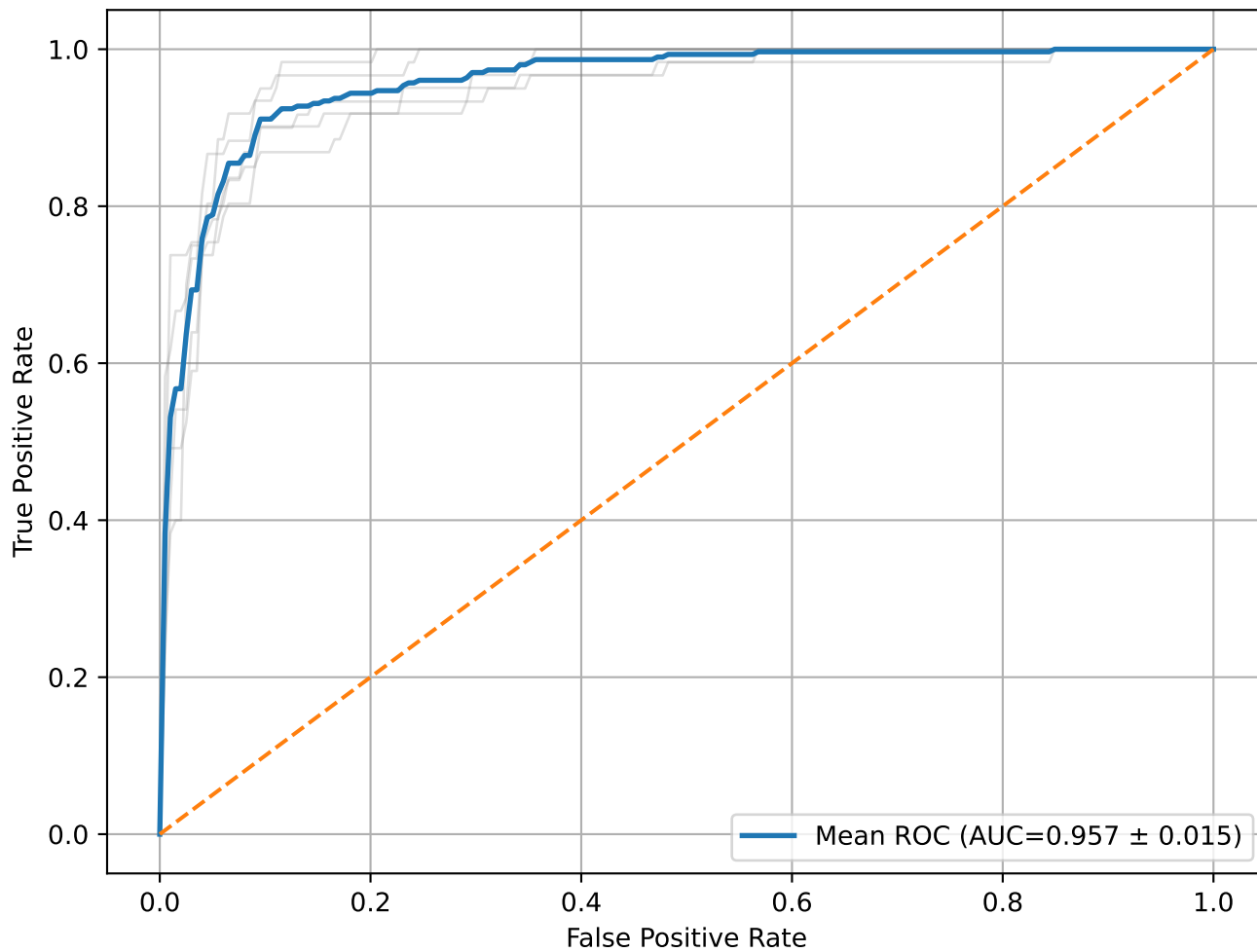
Confusion Matrix - LGB (last fold)



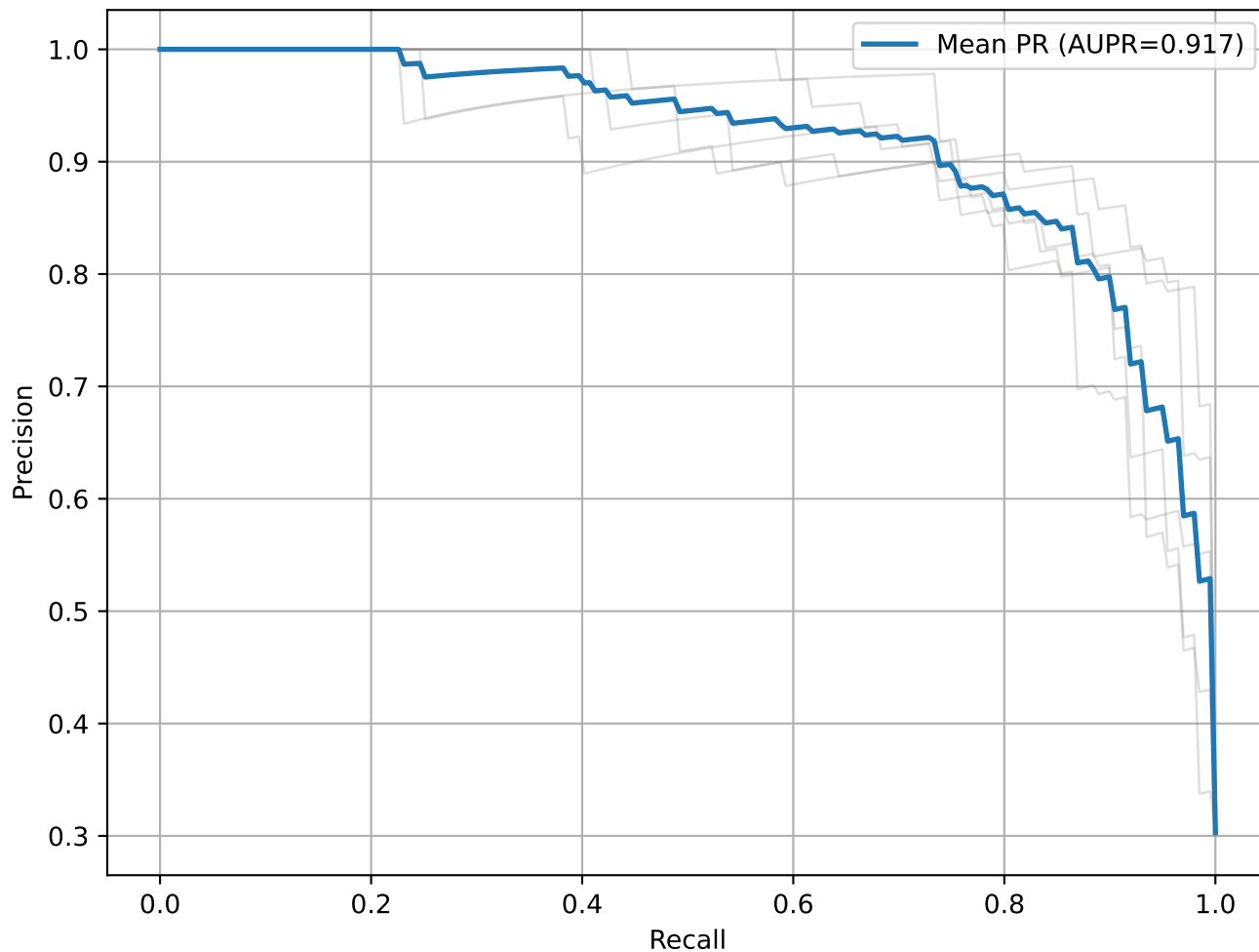
	precision	recall	f1-score	support
0	0.94	0.97	0.95	139
1	0.93	0.85	0.89	61
accuracy			0.94	200
macro avg	0.93	0.91	0.92	200
weighted avg	0.93	0.94	0.93	200

Classification Report - LGB (last fold)

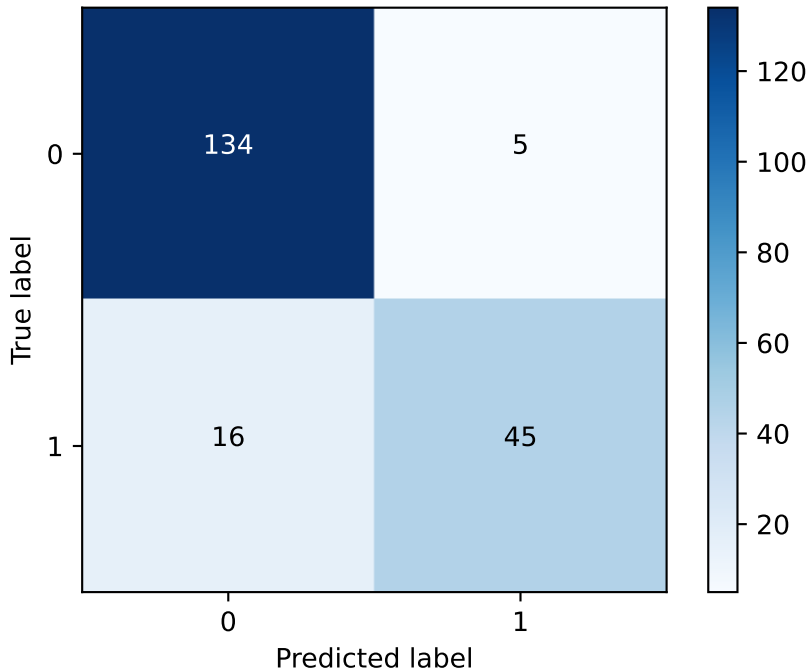
ROC - DNN



Precision-Recall - DNN



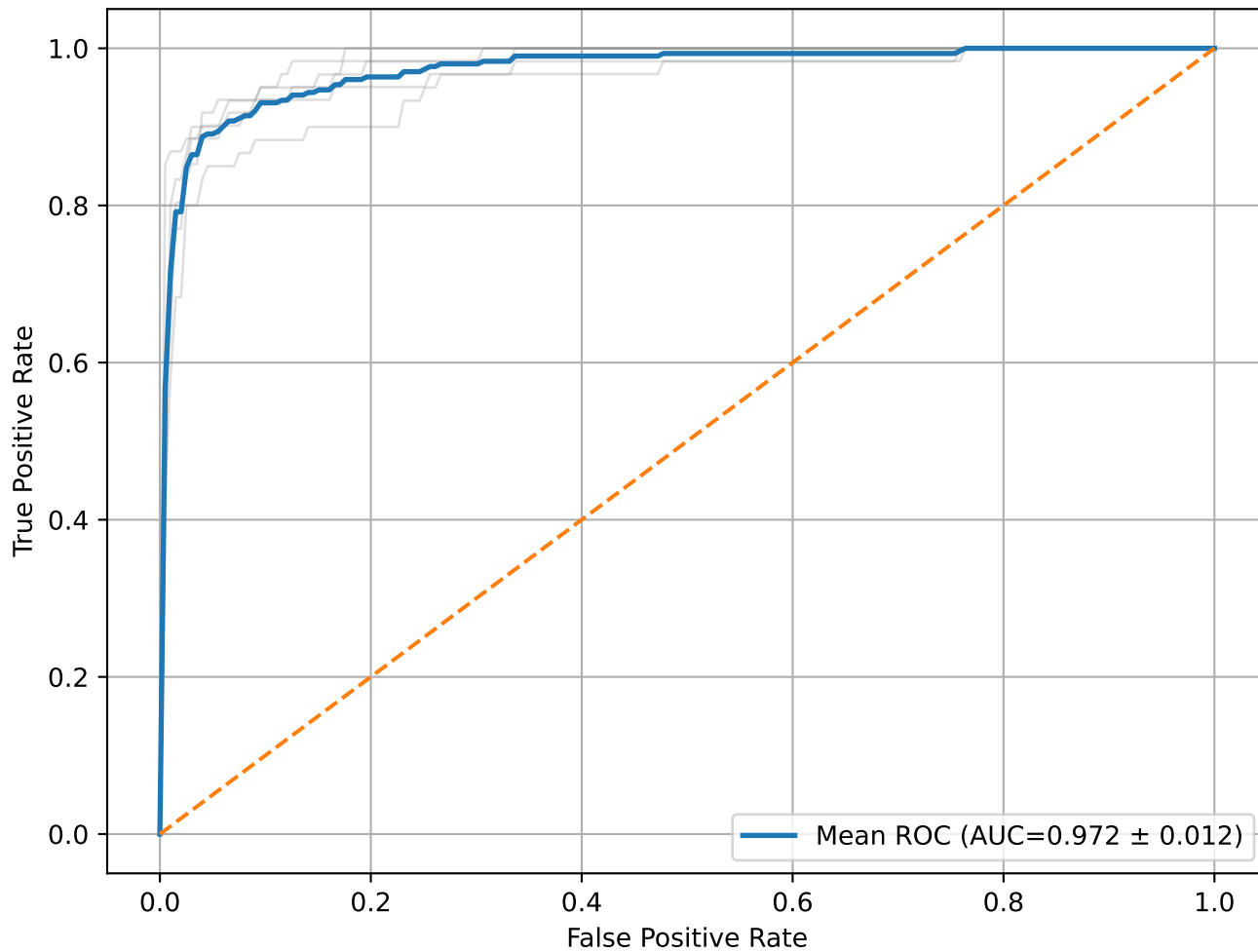
Confusion Matrix - DNN (last fold)



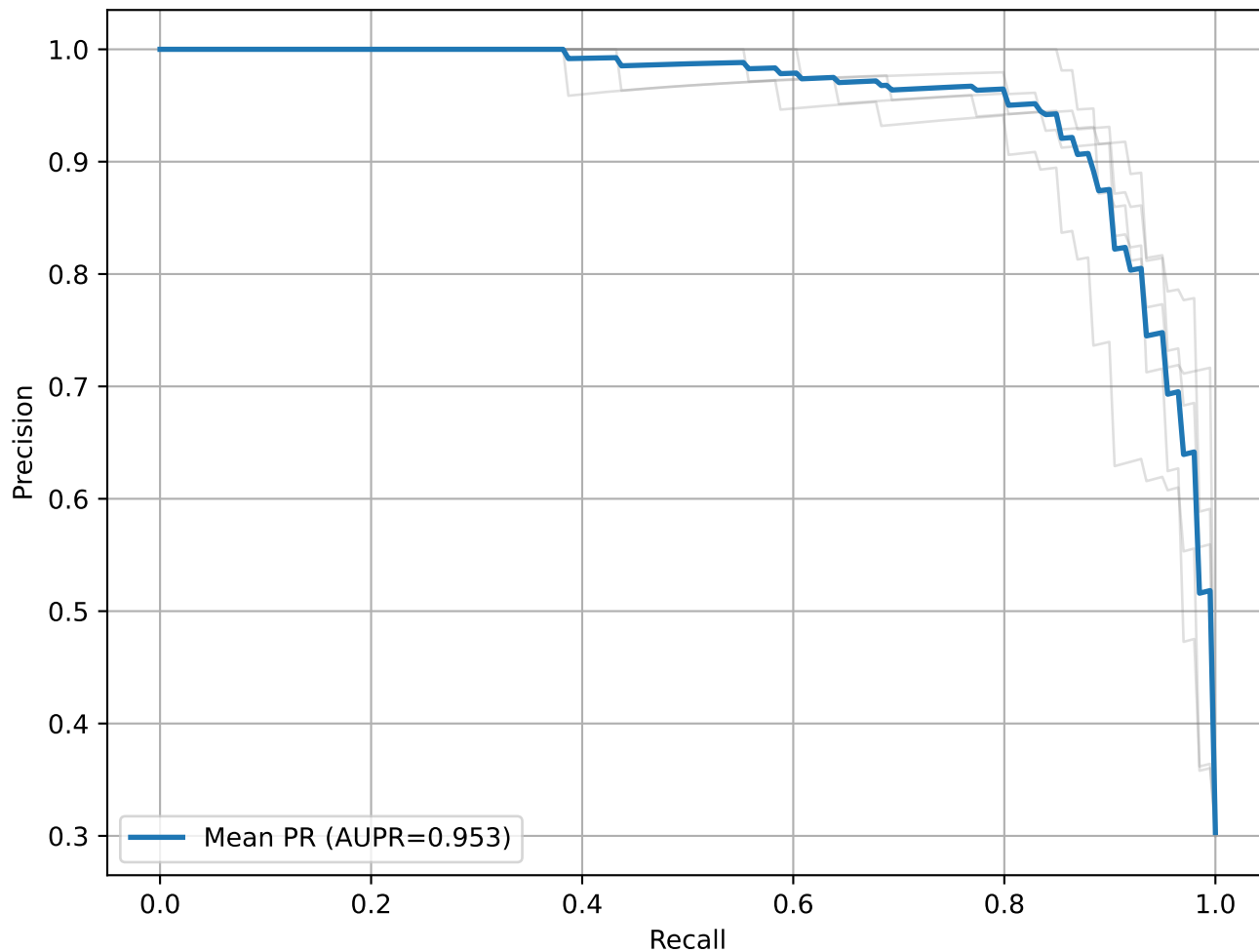
	precision	recall	f1-score	support
0	0.89	0.96	0.93	139
1	0.90	0.74	0.81	61
accuracy			0.90	200
macro avg	0.90	0.85	0.87	200
weighted avg	0.90	0.90	0.89	200

Classification Report - DNN (last fold)

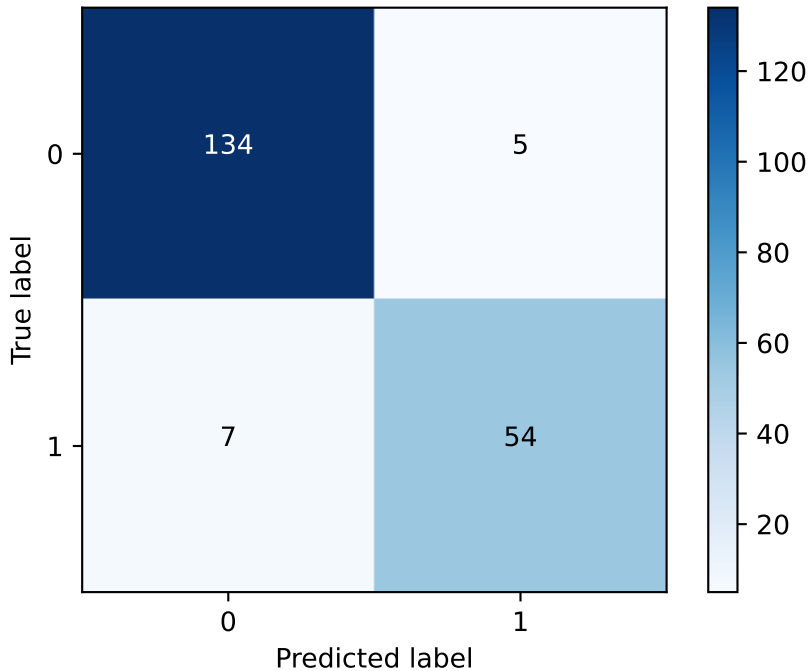
ROC - STACKING



Precision-Recall - STACKING



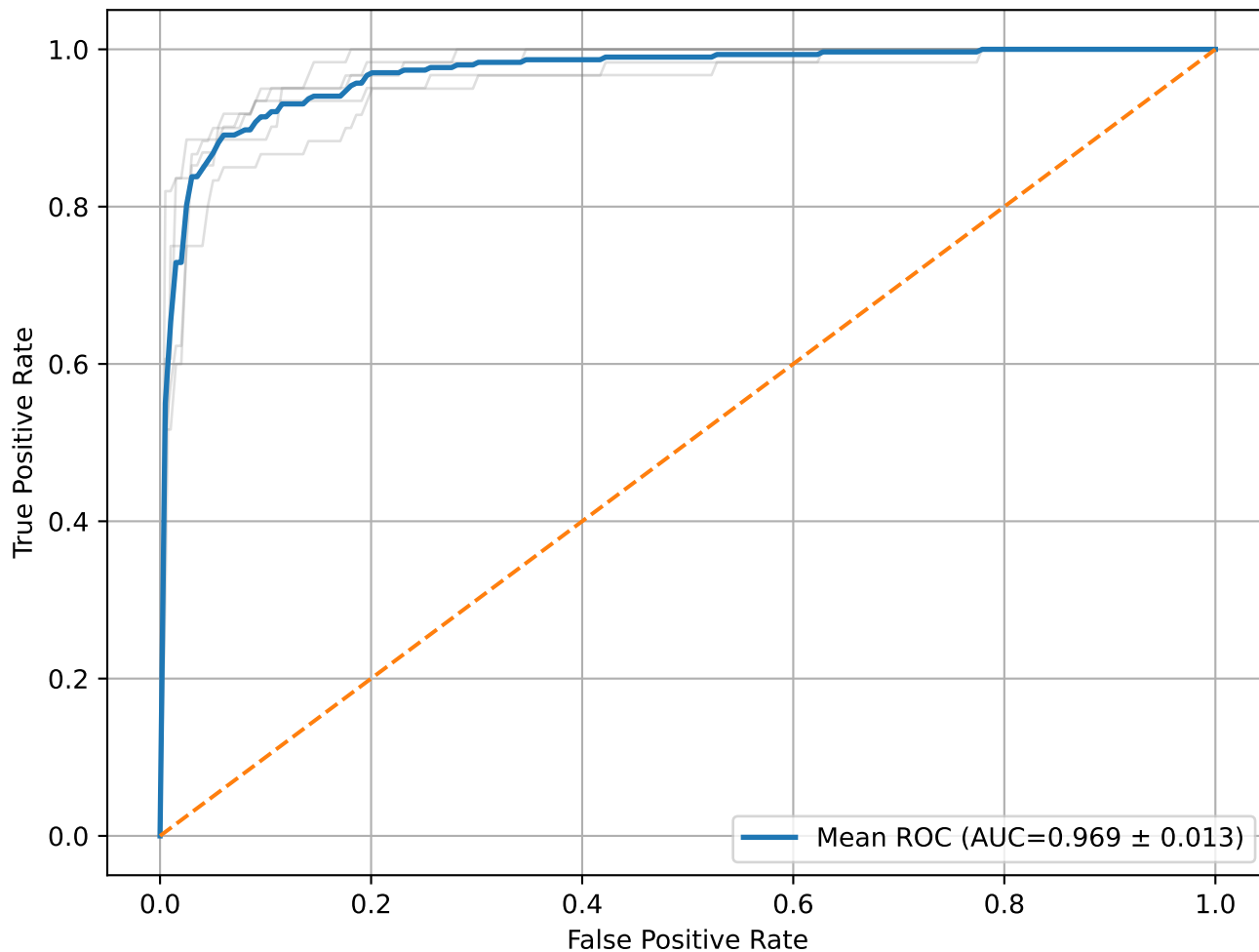
Confusion Matrix - STACKING (last fold)



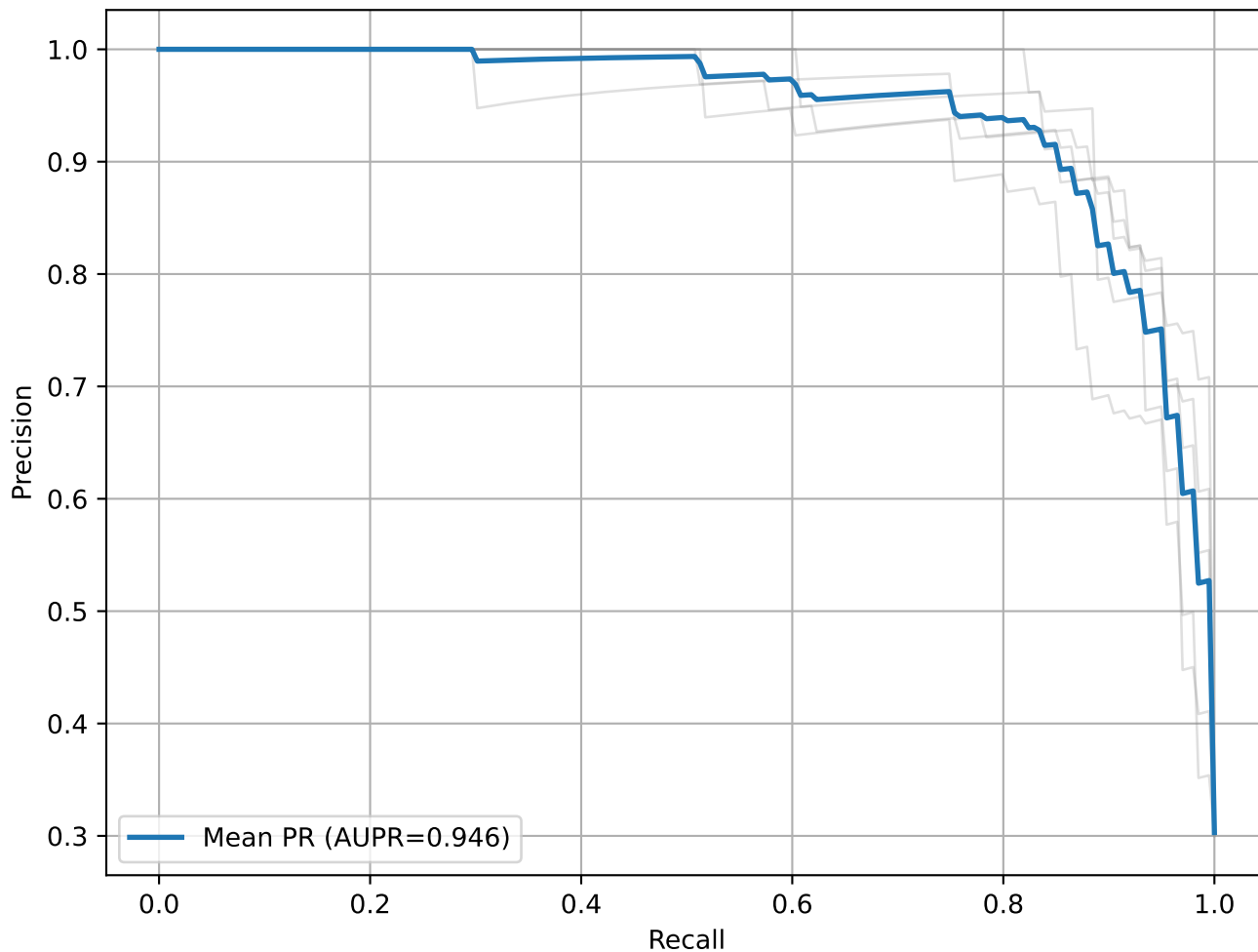
	precision	recall	f1-score	support
0	0.95	0.96	0.96	139
1	0.92	0.89	0.90	61
accuracy			0.94	200
macro avg	0.93	0.92	0.93	200
weighted avg	0.94	0.94	0.94	200

Classification Report - STACKING (last fold)

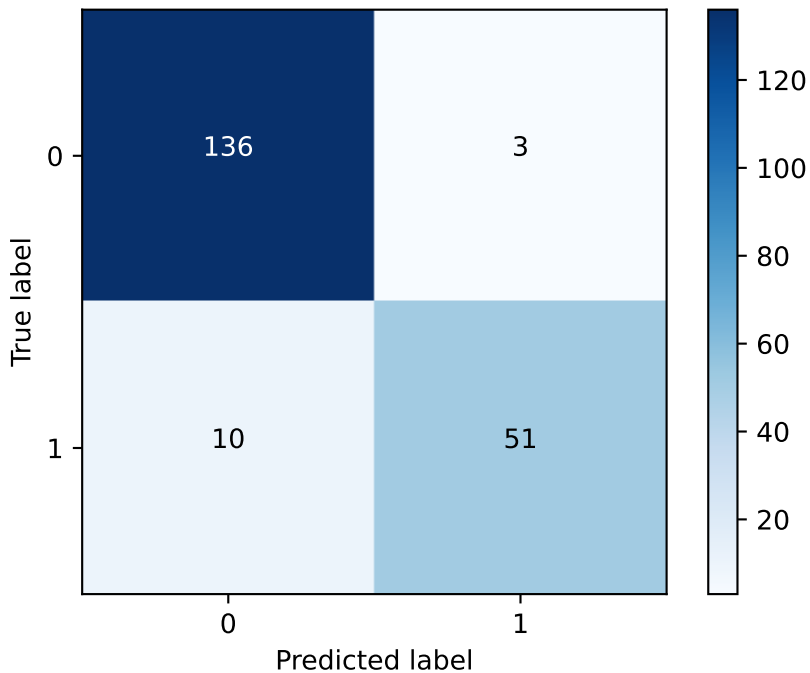
ROC - VOTING



Precision-Recall - VOTING



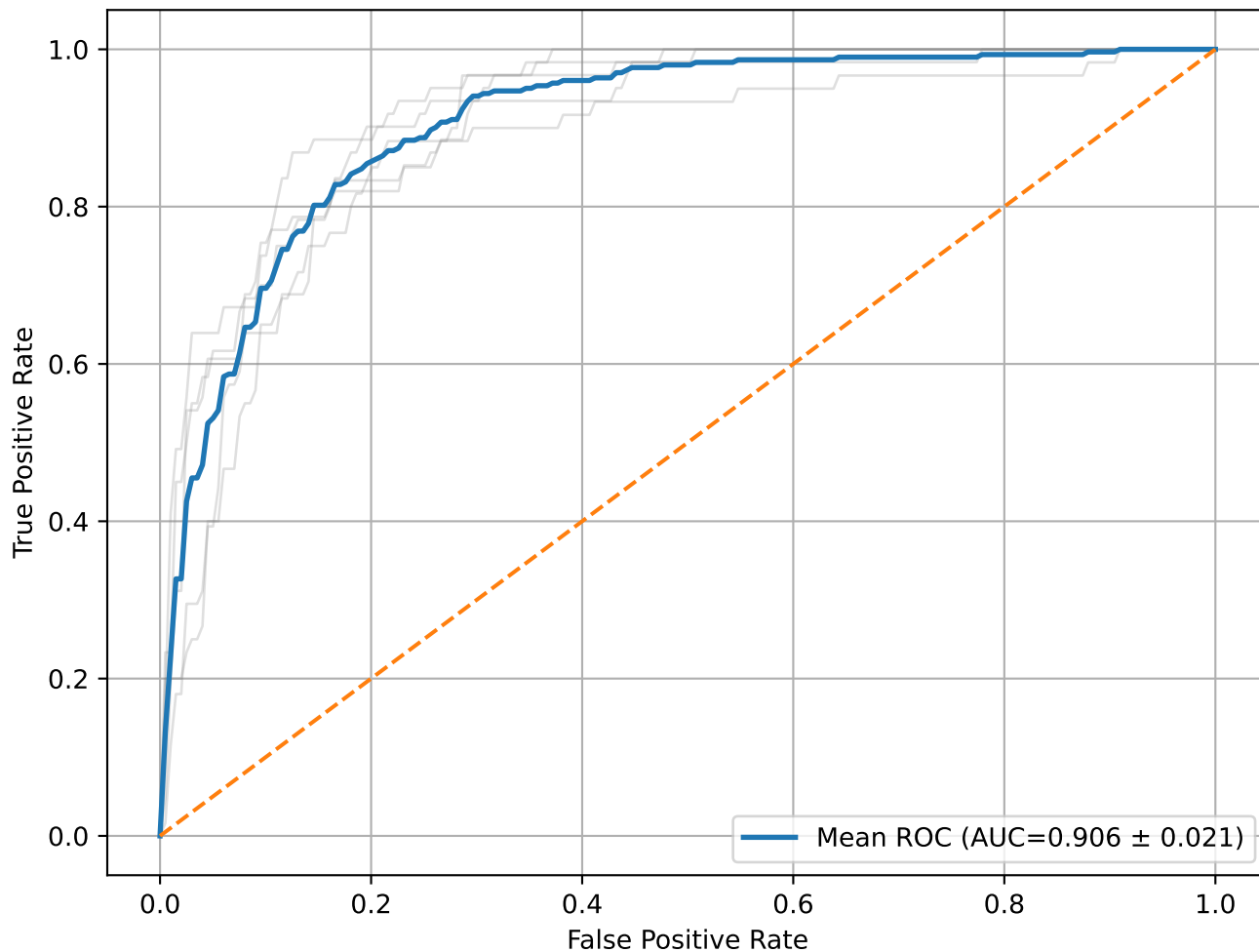
Confusion Matrix - VOTING (last fold)



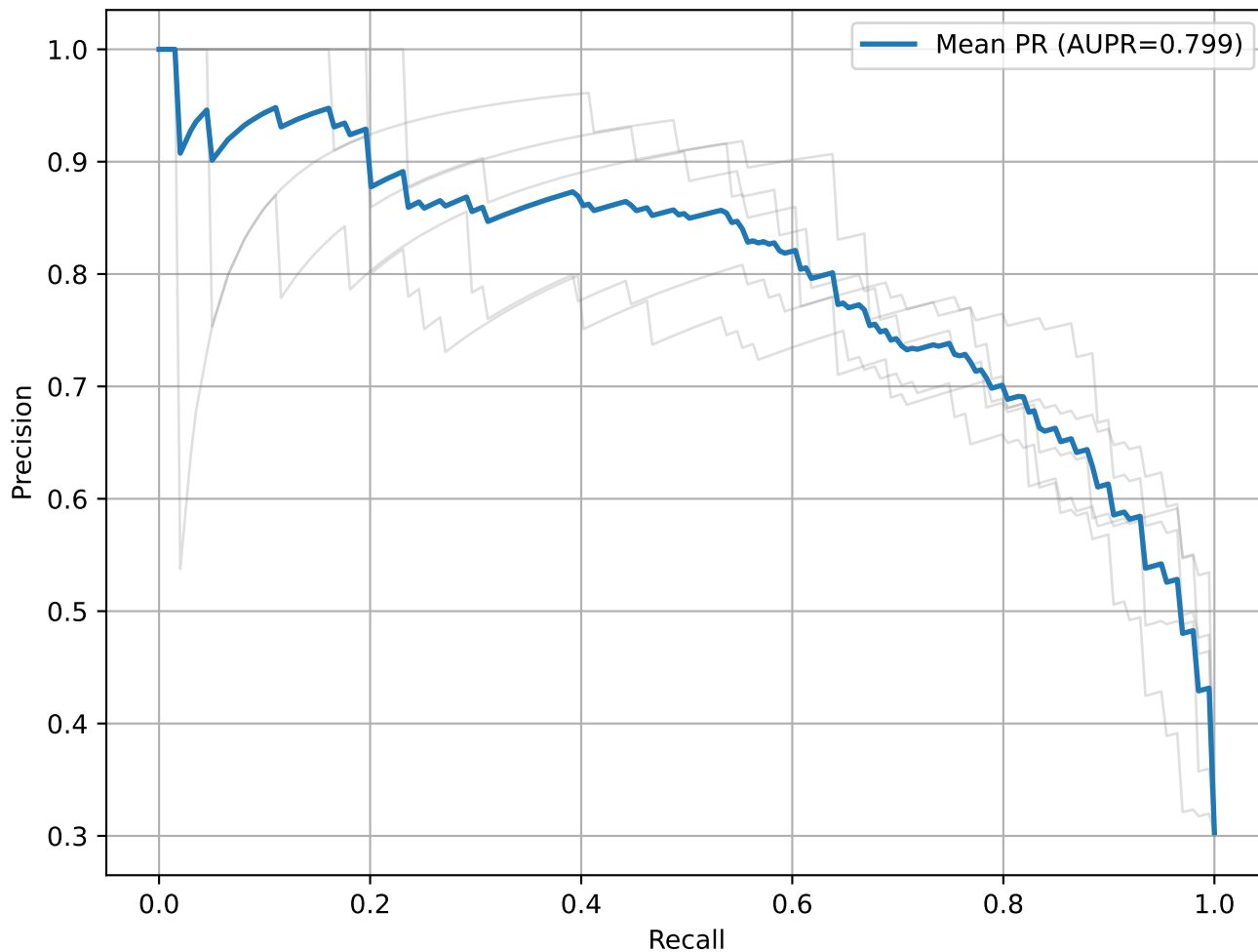
	precision	recall	f1-score	support
0	0.93	0.98	0.95	139
1	0.94	0.84	0.89	61
accuracy			0.94	200
macro avg	0.94	0.91	0.92	200
weighted avg	0.94	0.94	0.93	200

Classification Report - VOTING (last fold)

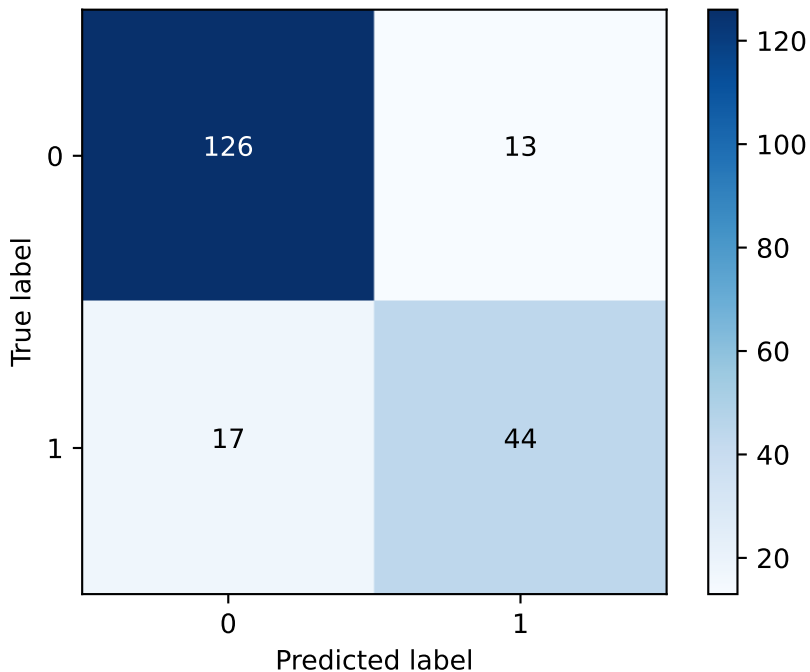
ROC - BAGGING



Precision-Recall - BAGGING



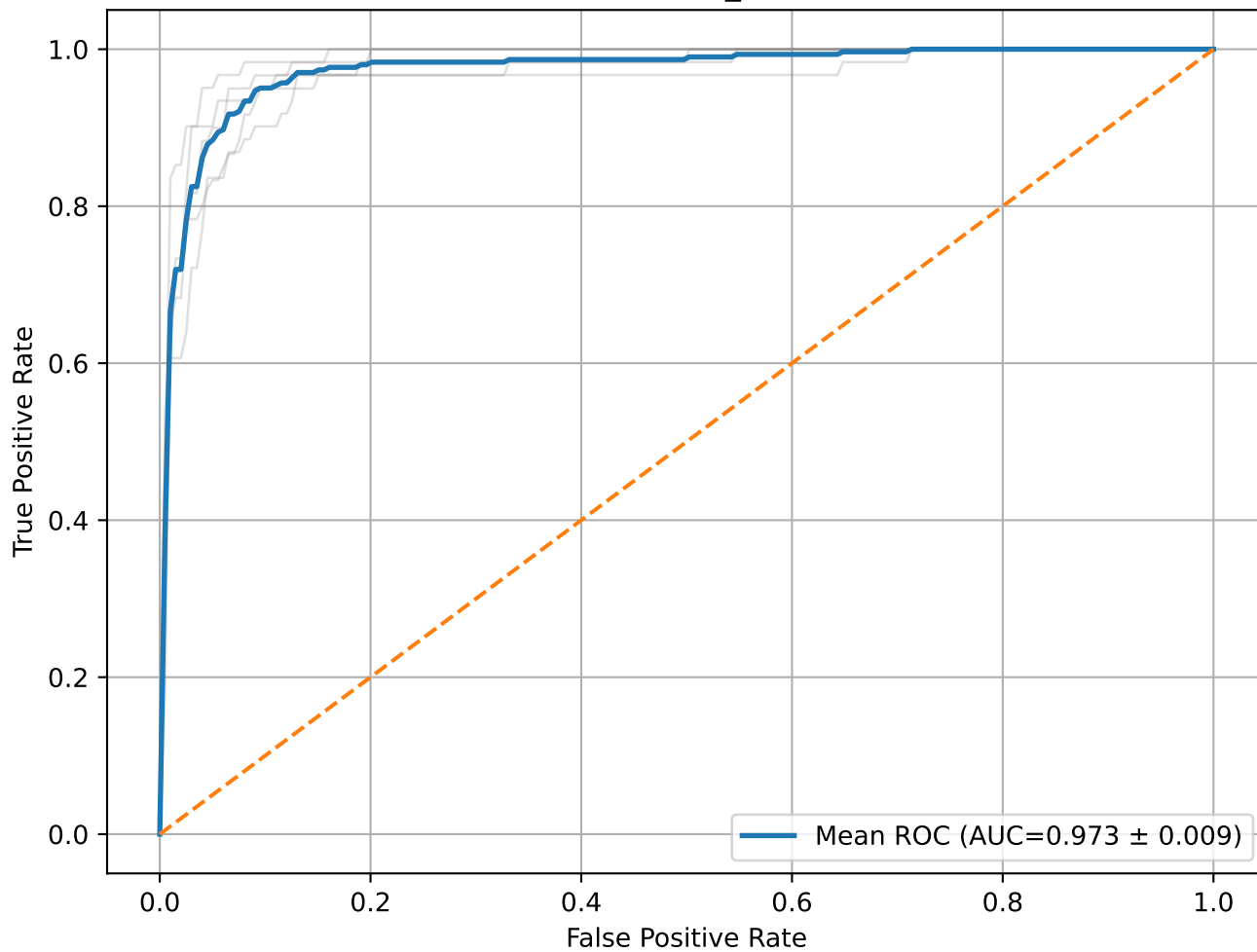
Confusion Matrix - BAGGING (last fold)



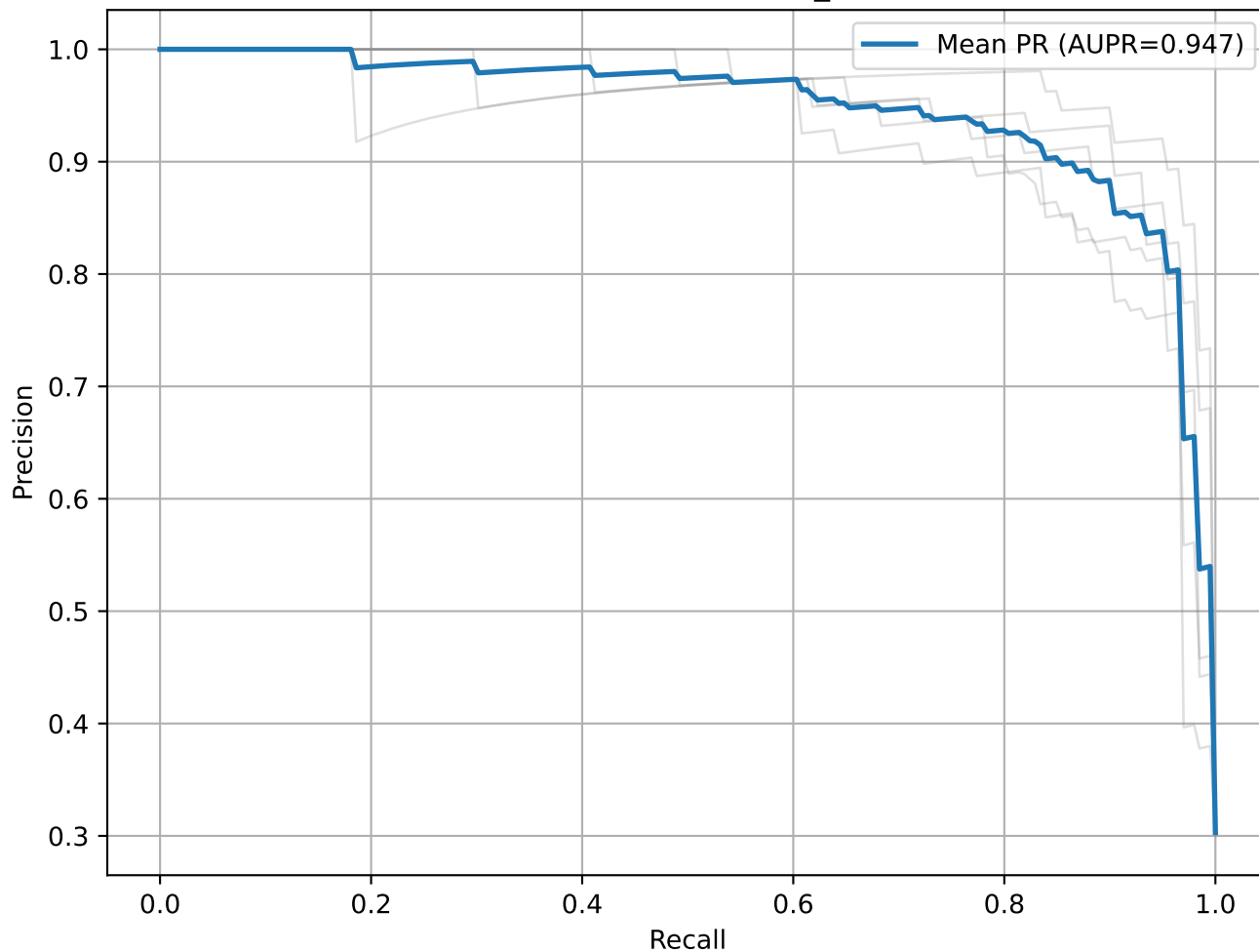
	precision	recall	f1-score	support
0	0.88	0.91	0.89	139
1	0.77	0.72	0.75	61
accuracy			0.85	200
macro avg	0.83	0.81	0.82	200
weighted avg	0.85	0.85	0.85	200

Classification Report - BAGGING (last fold)

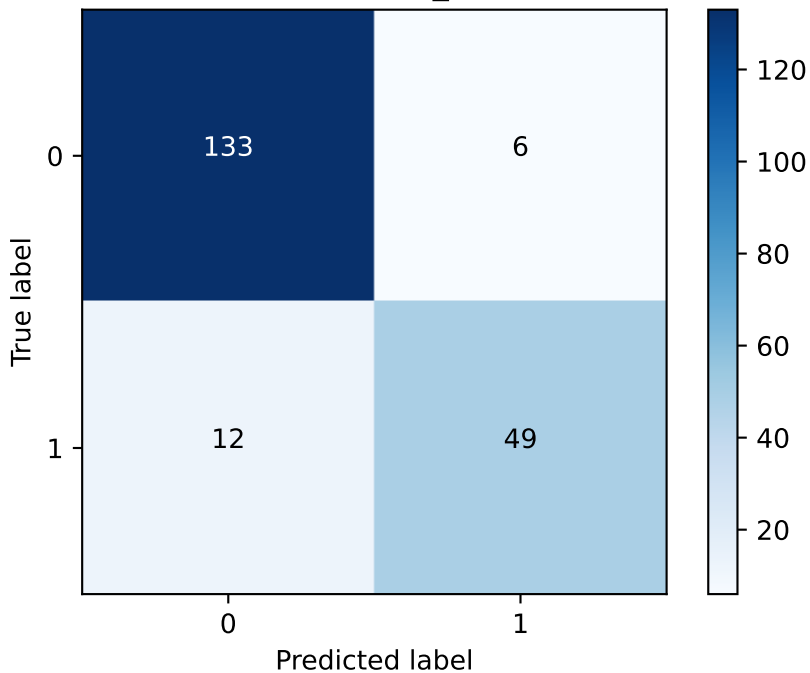
ROC - PCA_SVM



Precision-Recall - PCA_SVM



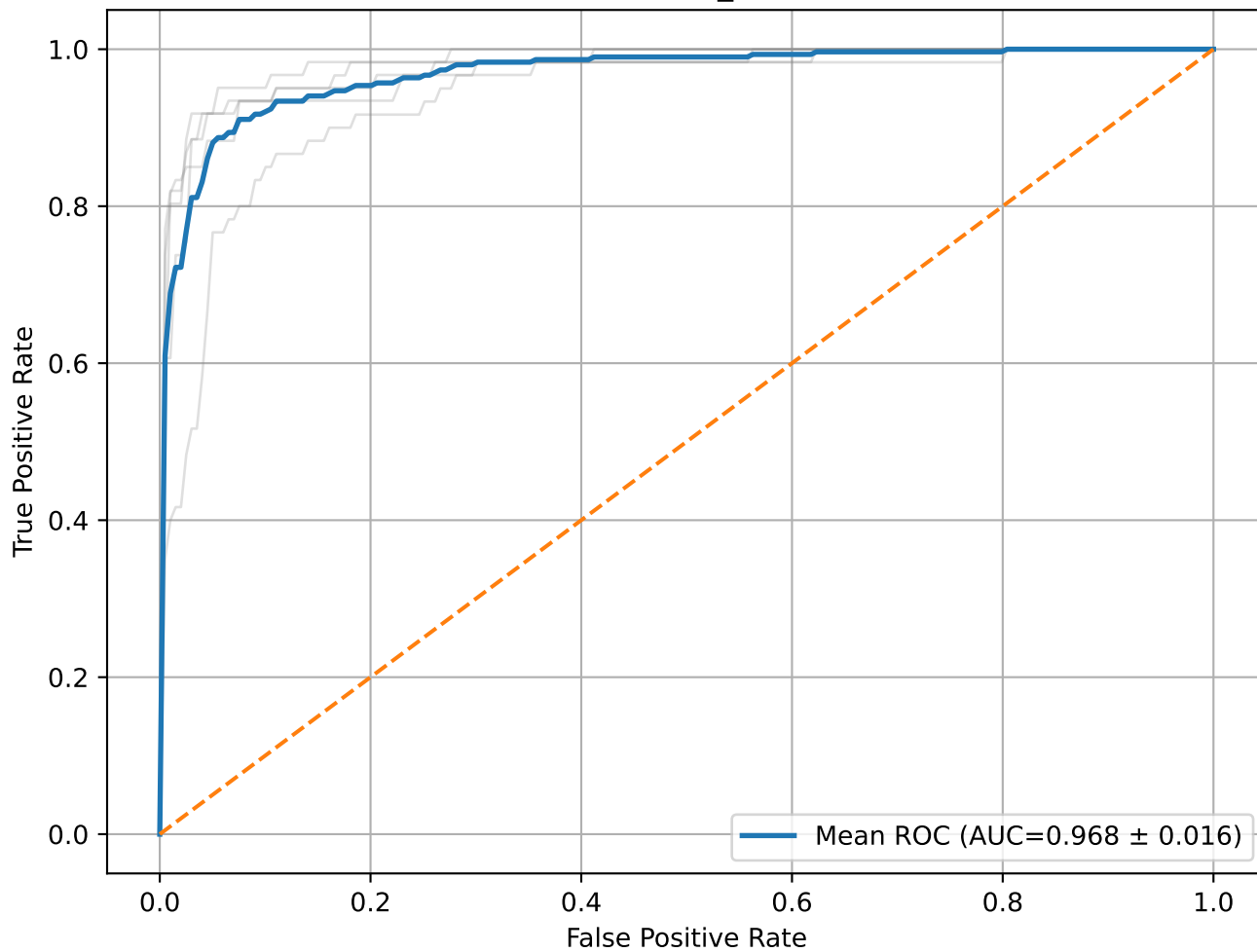
Confusion Matrix - PCA_SVM (last fold)



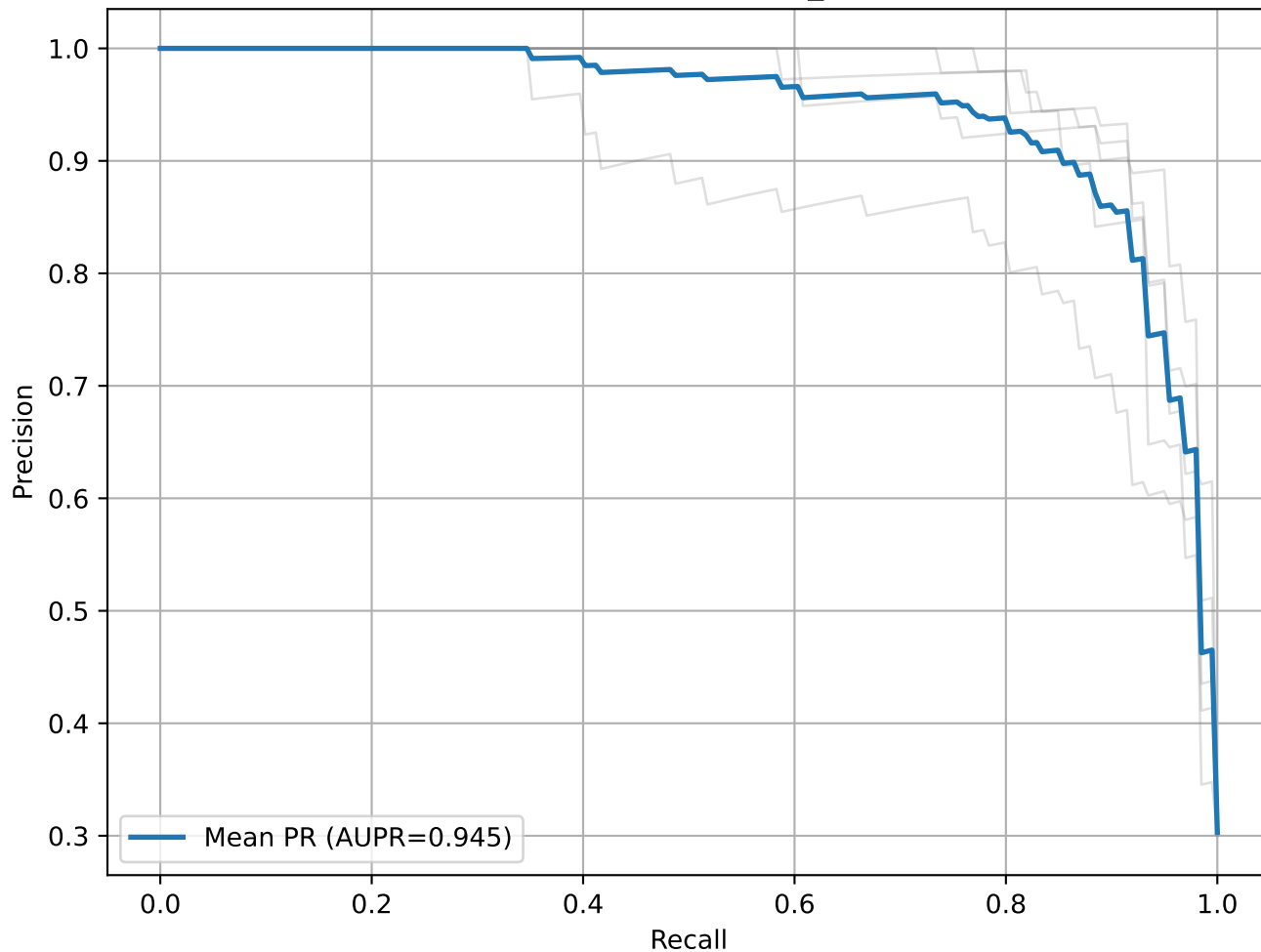
	precision	recall	f1-score	support
0	0.92	0.96	0.94	139
1	0.89	0.80	0.84	61
accuracy			0.91	200
macro avg	0.90	0.88	0.89	200
weighted avg	0.91	0.91	0.91	200

Classification Report - PCA SVM (last fold)

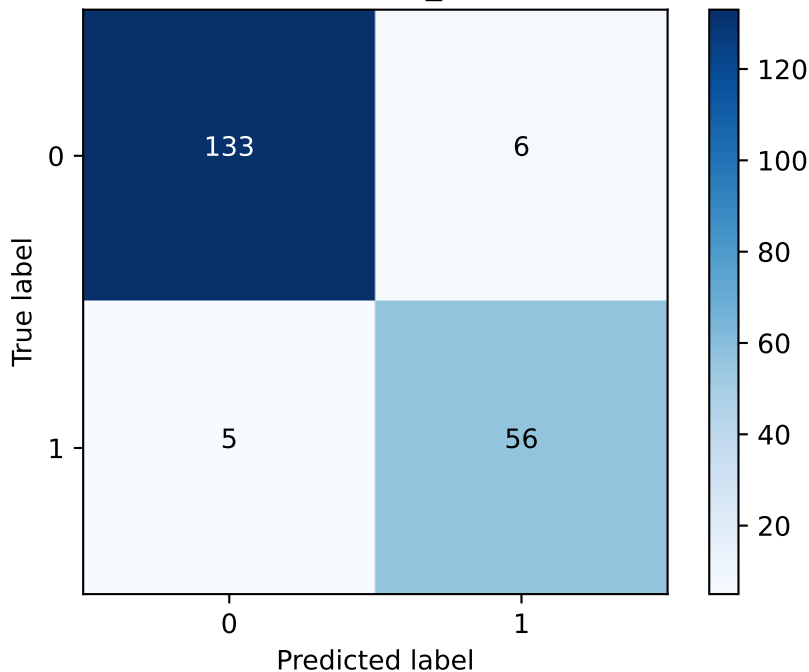
ROC - GA_XGB



Precision-Recall - GA_XGB



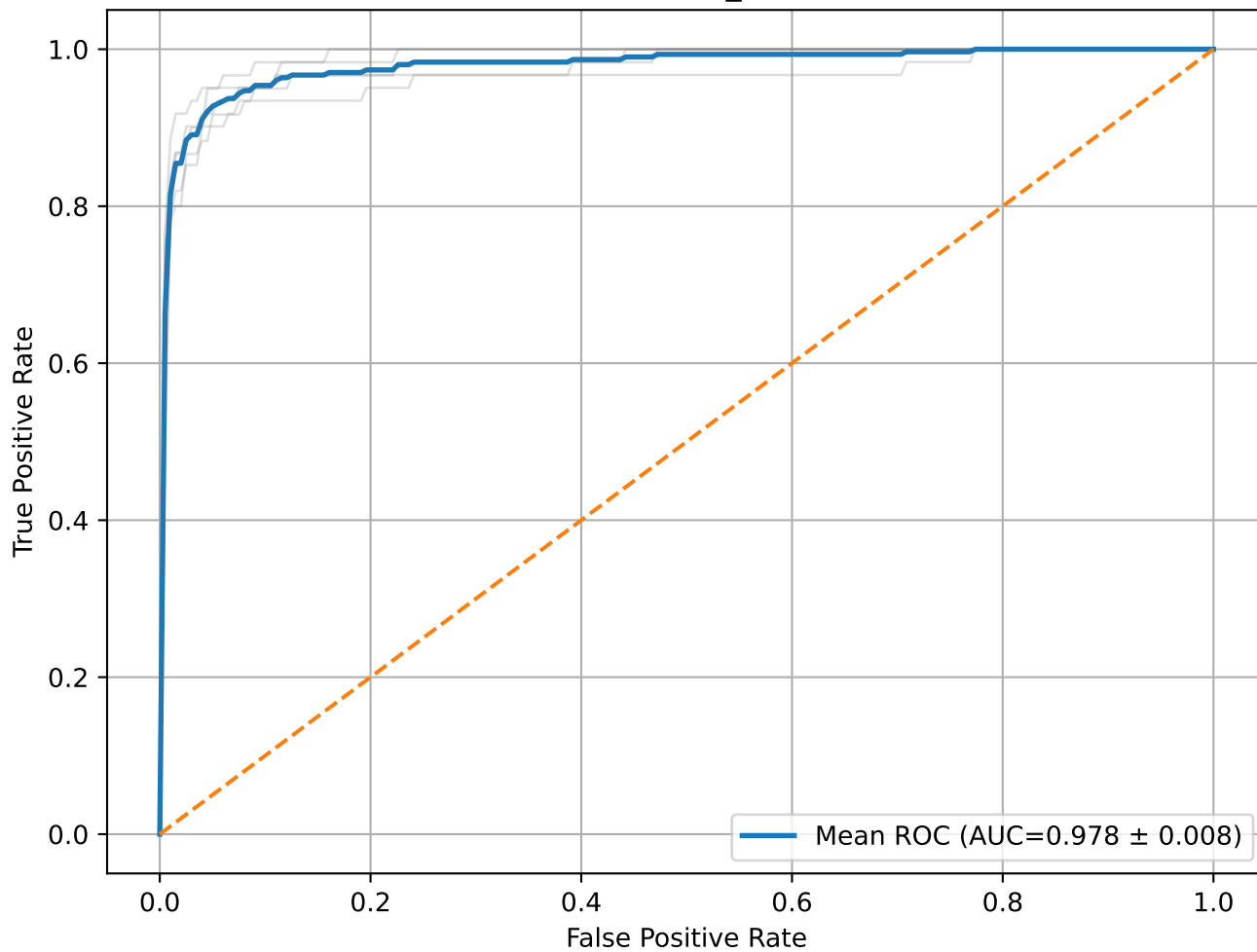
Confusion Matrix - GA_XGB (last fold)



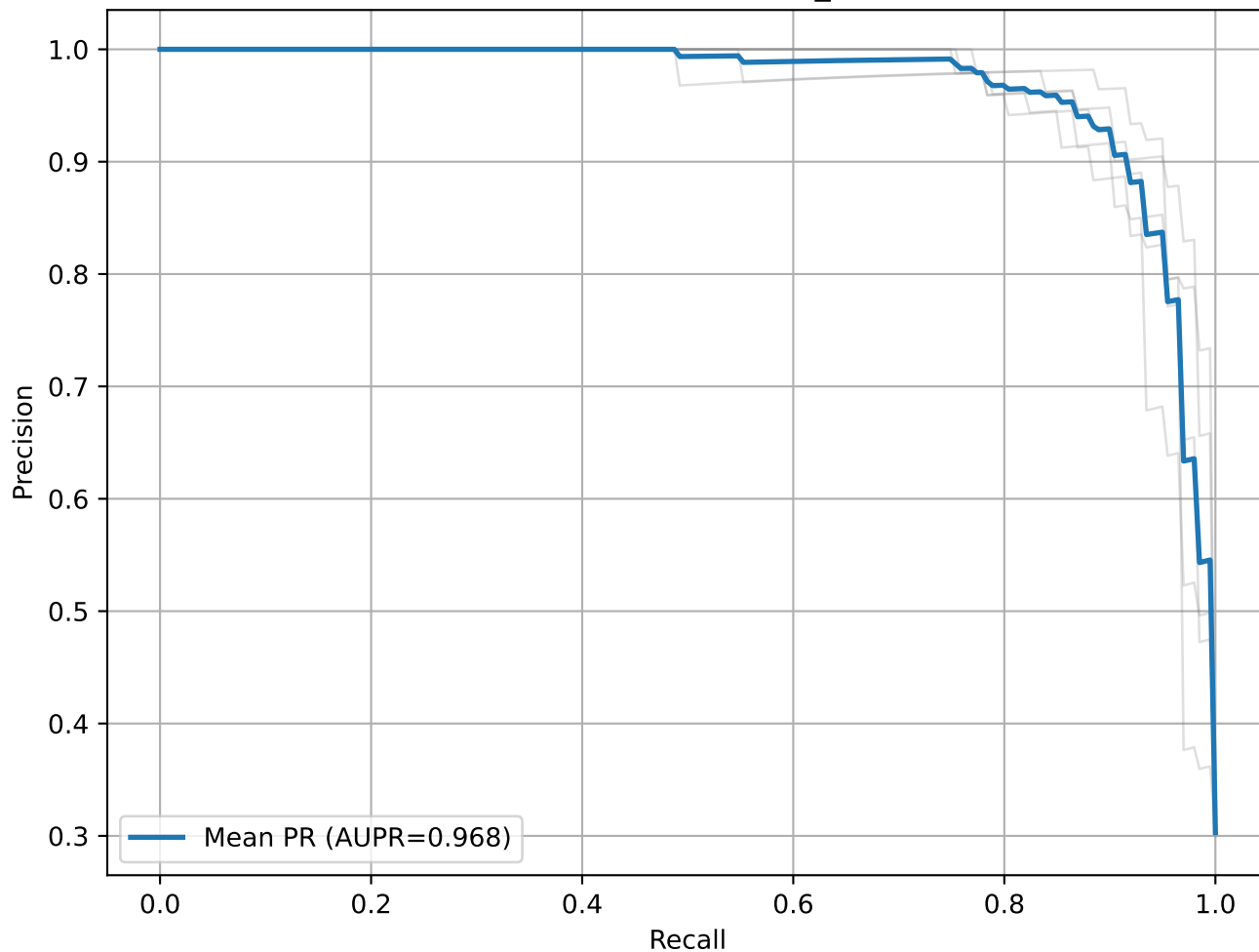
	precision	recall	f1-score	support
0	0.96	0.96	0.96	139
1	0.90	0.92	0.91	61
accuracy			0.94	200
macro avg	0.93	0.94	0.94	200
weighted avg	0.95	0.94	0.95	200

Classification Report - GA_XGB (last fold)

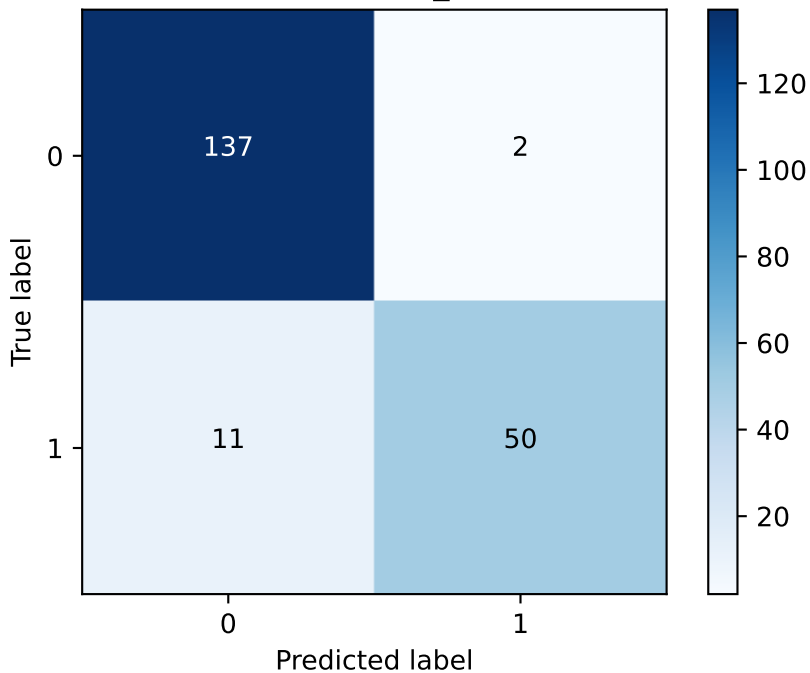
ROC - PSO_SVM



Precision-Recall - PSO_SVM



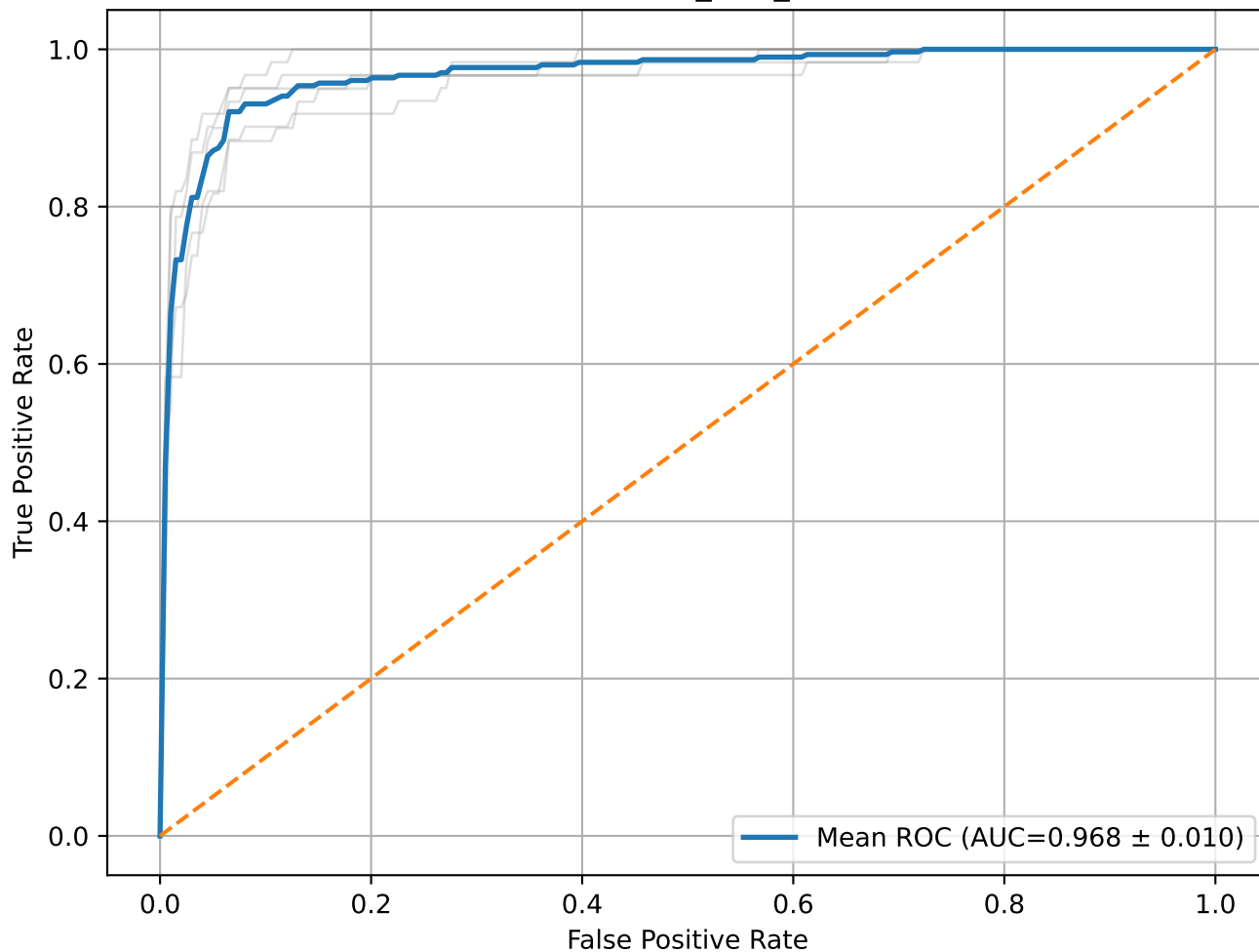
Confusion Matrix - PSO_SVM (last fold)



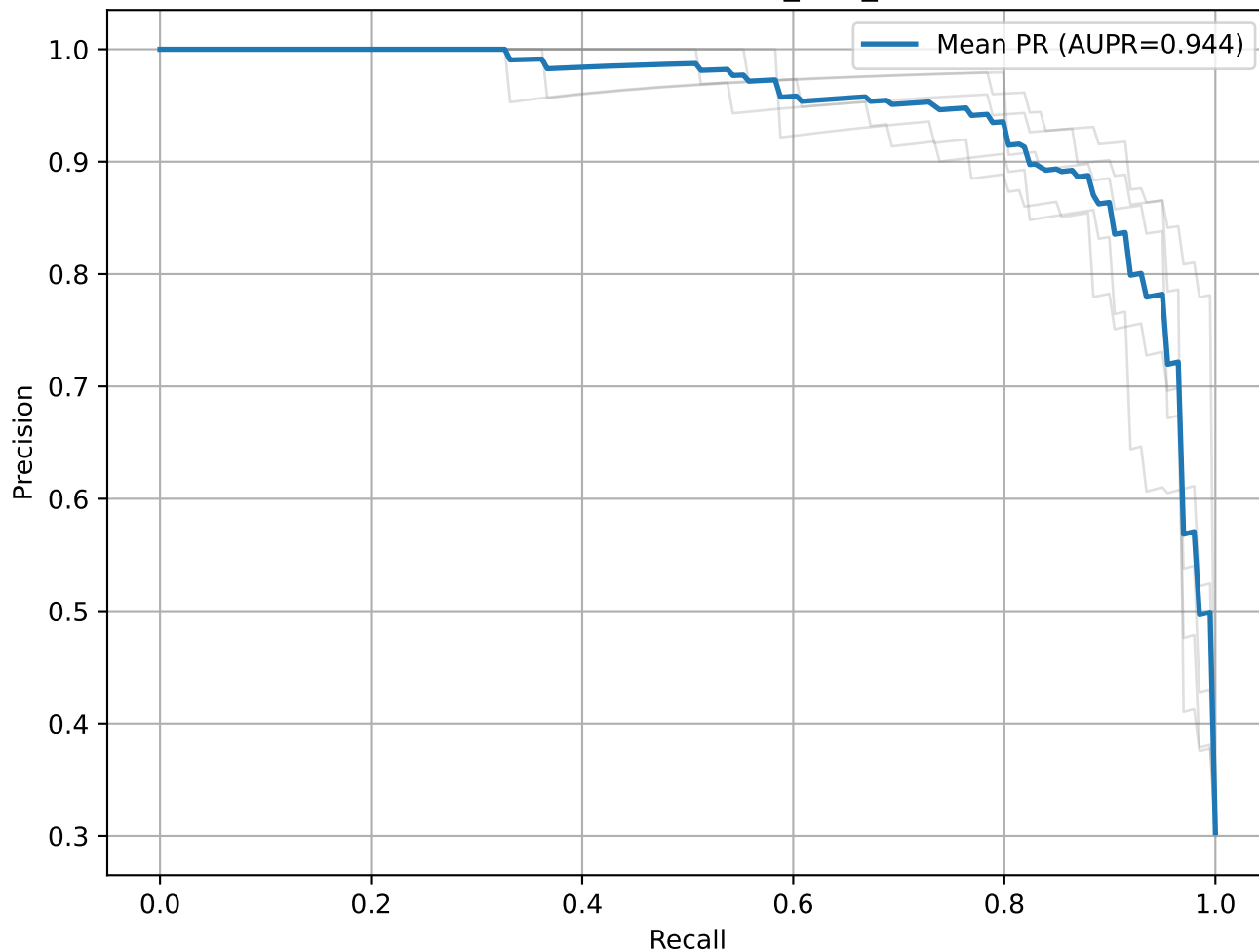
	precision	recall	f1-score	support
0	0.93	0.99	0.95	139
1	0.96	0.82	0.88	61
accuracy			0.94	200
macro avg	0.94	0.90	0.91	200
weighted avg	0.94	0.94	0.93	200

Classification Report - PSO SVM (last fold)

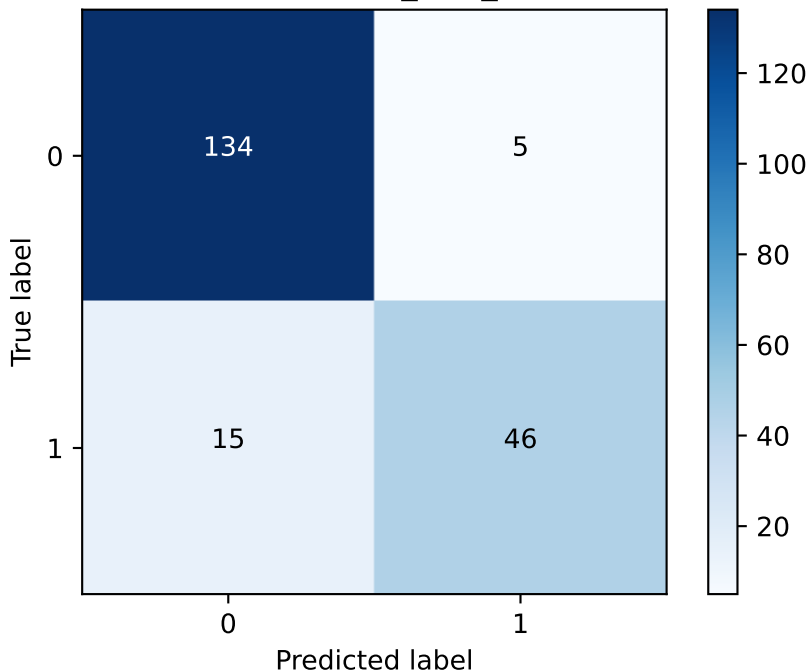
ROC - BAYES_OPT_SVMS



Precision-Recall - BAYES_OPT_SVMS



Confusion Matrix - BAYES_OPT_SVMS (last fold)



	precision	recall	f1-score	support
0	0.90	0.96	0.93	139
1	0.90	0.75	0.82	61
accuracy			0.90	200
weighted avg	0.90	0.86	0.88	200
weighted avg	0.90	0.90	0.90	200

Classification Report - BAYES_OPT_SVMS (last fold)

Model Comparison Table (sorted by mean ROC AUC)

model	mean_roc_auc	std_roc_auc	mean_aupr	mean_accuracy	mean_f1
pso_svm	0.980150	0.007931	0.968222	0.943	0.901431
lgb	0.977568	0.009426	0.961151	0.940	0.897689
rf	0.974338	0.013464	0.955441	0.925	0.862994
pca_svm	0.973920	0.009113	0.947238	0.931	0.880364
stacking	0.973364	0.012041	0.954011	0.940	0.897543
svm	0.971676	0.011248	0.945029	0.922	0.862598
xgb	0.971194	0.009856	0.949168	0.933	0.885994
voting	0.969975	0.012529	0.945990	0.932	0.881031
ga_xgb	0.969723	0.016179	0.945999	0.925	0.871115
bayes_opt_svms	0.968921	0.009748	0.944777	0.920	0.861411
dnn	0.958413	0.015275	0.917799	0.910	0.846343
logistic	0.908423	0.018803	0.801796	0.839	0.723316
bagging	0.906032	0.021107	0.799111	0.836	0.720052

Combined ROC - All Models

